Who Responds to Changes in the Market Risk Premia, and How?

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Abstract

Variation in market risk premia is reflected in the portfolio decisions of only a limited and identifiable subset of institutional investors. Using quarterly 13F holdings, I show that approximately 60% of institutions do not systematically adjust aggregate equity exposure as compensation for bearing market risk changes over time. Among those that do respond, exposure adjustments are highly heterogeneous in both sign and magnitude. To understand what drives these differences in exposure responses, I develop a simple framework in which investment horizons and short-run constraints determine investors' willingness and ability to adjust risk exposure. Consistent with the model's predictions, low-turnover and long-horizon investors, particularly pension funds, are significantly more likely to increase equity exposure when compensation for risk raises. These findings highlight the importance of a limited, identifiable set of investors who shift capital in response to changing risk compensation, while highlighting the lack of systematic exposure adjustment among majority of investors.

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Introduction

Expected equity returns vary over time. A large literature shows that business-cycle variables such as dividend yield, the default spread, and the term spread forecast future stock-market returns and capture variation in aggregate equity risk premia (Campbell & Shiller, 1988b; Cochrane, 2008a; Cochrane & Piazzesi, 2005a; Fama & French, 1989a). While the evidence on return predictability is extensive, much less is known about how this variation in expected returns is reflected in investors' portfolio decisions. Which institutions systematically adjust aggregate equity exposure when market risk premia change? Do investors respond in the same direction, or do some expand risk exposure while others reduce it? And what institutional characteristics explain these differences?

These questions are important because changes in expected returns are ultimately transmitted through investors' demand for risky assets. When compensation for bearing aggregate risk rises, some investors may be willing to absorb additional equity risk, while others may face funding constraints, benchmark concerns, regulatory limits, or short-run performance pressures that prevent them from doing so. Identifying which institutions adjust exposure, how much capital they move, and what drives these decisions is therefore central to understanding how aggregate risk is absorbed in financial markets.

This paper studies how U.S. institutional investors adjust aggregate equity exposure in response to changes in market risk premia. Using quarterly Form 13F holdings, I construct investor-level measures of active equity exposure and estimate, for each institution, its sensitivity to fluctuations in expected equity returns. Market risk premia are proxied using three variables widely used in the return-predictability literature: dividend yield, the default spread, and the term spread. Dividend yield captures variation in valuation and long-horizon discount rates (Campbell & Shiller, 1988b; Cochrane, 2008a), the default spread reflects compensation for bearing credit and macroeconomic risk (Chen et al., 1986a; Fama & French, 1989a), and the term spread captures variation in business-cycle conditions and duration-related risks (Cochrane & Piazzesi, 2005a; Fama & French, 1989a). I also construct a fitted

expected-return factor from these predictors. The resulting institution-specific sensitivities, which I refer to as *macro betas*, provide a reduced-form measure of how investors adjust aggregate equity exposure across business-cycle states.

The first finding is that systematic exposure responses to changes in market risk premia are far from universal. Roughly 60% of institutions exhibit no statistically significant response to any of the three risk-premium proxies. This does not imply that these investors are absent from equity markets or that they are passive investors. Rather, their aggregate equity exposure does not systematically vary with the expected-return measures studied in this paper. The evidence suggests that variation in market risk premia is translated into portfolio reallocations by only a limited identifiable subset of institutional investors.

The second finding is that exposure responses are highly heterogeneous among the institutions that do respond. Using the fitted expected-return specification, I find that following a one-percentage-point increase in annual expected equity returns, approximately 15% of institutions increase equity exposure by an average of 2% of assets under management, while roughly 5% reduce exposure by an average of 7% of assets under management. Thus, investors observing the same change in expected returns often respond in sharply different ways.

This heterogeneity is systematically related to investor characteristics. A broad class of theories suggests that institutions differ in investment horizons, funding stability, balance-sheet constraints, benchmark incentives, and organizational objectives (Brunnermeier & Sannikov, 2014a; Campbell & Viceira, 2002; He & Krishnamurthy, 2013a; Shleifer & Vishny, 1997a). Consistent with these ideas, I find that low-turnover investors exhibit substantially more positive macro betas than high-turnover investors. Pension funds are among the institutions most likely to increase equity exposure when valuation and credit-related risk premia rise, whereas banks and investment advisors respond more strongly to variation in the term structure. These patterns suggest that differences in investment horizon and institutional constraints are important determinants of how investors translate changes in expected re-

turns into portfolio decisions.

To interpret these findings, I develop a simple framework in which investors differ in investment horizons and short-run constraints. In the model, long-horizon institutions place greater weight on persistent variation in expected returns and therefore increase equity exposure when compensation for bearing aggregate risk rises. By contrast, institutions facing stronger short-run constraints, such as interim-performance concerns, funding pressures, redemption risk, or regulatory restrictions, may respond less aggressively or even reduce exposure when risk premia rise. The model generates two predictions consistent with the empirical evidence: exposure responses should be heterogeneous in both sign and magnitude, and long-horizon investors should exhibit more positive exposure sensitivities than short-horizon investors.

This paper contributes to the intersection of the return-predictability, intermediary asset-pricing, and demand-based asset-pricing literatures. A large literature shows that expected returns vary over time and can be forecast using business-cycle variables such as dividend yield, the default spread, and the term spread (Campbell & Shiller, 1988a; Cochrane, 2008b; Fama & French, 1989b). A related literature emphasizes that heterogeneity in institutions' funding conditions, balance-sheet constraints, and risk-bearing capacity plays an important role in determining asset prices and expected returns (Brunnermeier & Sannikov, 2014b; He & Krishnamurthy, 2013b; Shleifer & Vishny, 1997b). More recently, the demand-based asset-pricing literature highlights the importance of heterogeneous investor demand in shaping asset prices (Koijen & Yogo, 2019). I connect these literatures by identifying which institutional investors adjust equity demand when expected returns change and by documenting substantial heterogeneity in those responses. I show that variation in market risk premia is reflected in the portfolio decisions of only a limited and identifiable subset of institutions, particularly long-horizon investors. These findings provide new evidence on how fluctuations in investment opportunities are transmitted into asset demand and, ultimately, asset prices.

The remainder of the paper proceeds as follows. Section 1 documents the predictive content of dividend yield, the default spread, and the term spread for future market returns. Section 2 estimates institution-level exposure sensitivities to market risk premia and documents heterogeneity across investors. Section 3 examines whether observable investor characteristics explain this heterogeneity. Section 4 develops a simple framework that rationalizes the empirical findings. Section 5 concludes.

1 Market Risk Premia

The first step in studying how institutions respond to variation in market risk premia is to measure those risk premia empirically. Expected returns are not directly observable and therefore must be inferred from variables that forecast future returns. A large asset-pricing literature has shown that several business-cycle indicators contain information about time-varying expected stock returns. Following this literature, I use three standard predictors of aggregate equity returns: dividend yield, the default spread, and the term spread.

The dividend yield captures variation in valuation and long-horizon discount rates (Campbell & Shiller, 1988a; Cochrane, 2008b). The default spread measures compensation for bearing credit and macroeconomic risk and tends to widen when financial conditions deteriorate (Chen et al., 1986b; Fama & French, 1989b). The term spread reflects variation in business-cycle conditions and duration-related risk, with a steeper yield curve generally associated with higher future excess returns (Cochrane & Piazzesi, 2005b; Fama & French, 1989b). Together, these variables provide a parsimonious and widely used characterization of time-varying market risk premia and serve as the state variables throughout the paper.

Before turning to investor-level responses, it is useful to verify that these variables continue to contain information about future market returns over the sample period studied in this paper. I therefore begin by examining their time-series properties and predictive power for aggregate equity returns.

Table 1 provides an initial overview of the three expected-return indicators. Panel A reports summary statistics over 1980–2024. Dividend yield is relatively stable and exhibits low volatility, the default premium shows episodic but pronounced spikes around recessionary periods, and the term spread displays the largest variation—ranging from deep inversions to steep positive slopes—consistent with its role as a forward-looking measure of monetary and business-cycle conditions. Market returns exhibit the expected high volatility, while investor-level equity adjustments are centered near zero but display substantial dispersion.

Panel B presents the correlation structure. The three indicators are only moderately cor-

related with one another, indicating that they capture distinct dimensions of macroeconomic and financial conditions. Their weak contemporaneous correlation with market returns is consistent with their interpretation as predictors of future, rather than current, expected returns.

Figure 7 illustrates the time-series behavior of the three business-cycle indicators. Dividend yield exhibits a pronounced secular decline, rising to nearly 6% in the early 1980s before falling to around 1% in the early 2000s and stabilizing thereafter. In contrast, the default spread shows little long-run trend but displays sharp cyclical spikes, most notably during periods of financial stress such as the 2008 crisis. The term spread fluctuates substantially over time, reflecting changing monetary conditions and business-cycle expectations, with alternating periods of steep and inverted yield curves.

1.1 Predictability of aggregate equity returns

To assess whether these indicators contain information about expected returns over the sample relevant for the investor-level analysis, Table 2 reports predictive regressions of cumulative market returns at horizons of one quarter, one year, and five years. Each predictor is examined individually and jointly.

The results confirm that the indicators remain meaningful signals of expected returns. The default premium is the strongest and most robust predictor. In univariate regressions, a one-percentage-point increase in the default premium predicts a 1.3 percentage-point increase in next-quarter returns, a 4.2 percentage-point increase over the next year, and a 24 percentage-point increase over the next five years. These effects remain large and statistically significant in the multivariate specification. The associated R^2 values rise from roughly 1% at the quarterly horizon to 16% at the five-year horizon, indicating that the default premium captures a substantial share of long-horizon variation in expected returns.

The term spread also forecasts returns, but primarily at long horizons. In univariate regressions, it loads strongly on five-year returns, consistent with its interpretation as a slow-

moving business-cycle indicator. Its predictive power weakens once the default premium is included jointly, suggesting that the two variables capture overlapping components of long-horizon discount-rate variation. The R^2 values mirror this pattern: the term spread explains little at short horizons but nearly 15% of five-year returns in univariate regressions.

Dividend yield shows limited predictive power in univariate regressions over this sample, but becomes statistically significant at the five-year horizon once the default premium and term spread are included. This pattern indicates that dividend yield contains long-horizon discount-rate information that is masked in univariate specifications but revealed when controlling for other business-cycle variables. Appendix A shows that its weak univariate performance reflects substantial time-variation in predictive content rather than an absence of information. More specifically, I split the sample into pre- and post-2000 periods and find that dividend yield exhibits no predictability prior to 2000, whereas in the post-2000 period it predicts returns across both short and long horizons.

Taken together, the coefficients and associated R^2 values reveal that the predictive content of these business-cycle variables is both economically meaningful and strongly horizon-dependent, with distinct patterns across indicators. The default premium emerges as the most robust predictor, with sizable effects at both short and long horizons and sharply increasing explanatory power over longer horizons. In contrast, the term spread primarily forecasts returns at long horizons, with little predictive power in the short run, consistent with its role as a slow-moving business-cycle indicator. Dividend yield exhibits a different pattern: while its unconditional predictive power is limited, particularly prior to 2000, it becomes informative at long horizons—and, in the post-2000 period, across both short and long horizons—once other state variables are controlled for. These differences suggest that expected-return signals are not only time-varying but also horizon-specific, raising the possibility that investors with different investment horizons may respond differentially to distinct components of aggregate risk premia.

Table 1 about here

Figure 7 about here

Table 2 about here

2 Investors Equity Exposure Dynamics

This section describes institutional investor data, the construction of investor-level equity exposure, the estimation of investor-specific macro betas, and the main empirical patterns that emerge from these estimates. The goal is to characterize how investors adjust their portfolios in response to movements in expected aggregate equity returns and how heterogeneous they are.

The analysis is based on quarterly holdings reported in Form 13F filings, obtained from the Thomson Reuters Institutional Holdings (s34) database. Institutions with more than $100 million in Section 13(f) securities are required to report their long equity positions each quarter. These filings cover a broad set of institutional investors, including banks, insurance companies, investment advisors, mutual funds, pension funds, and other asset managers. Short positions, cash, and bonds are not reported.

Holdings are merged with CRSP data on prices and shares outstanding using CUSIP identifiers. For each stock, household-sector holdings are computed as the difference between shares outstanding and the total shares held by 13F institutions, capturing direct household ownership and smaller institutions below the reporting threshold. The final sample includes all institutions with at least 60 quarters of observable history, yielding 2,087 investors with an average time series of roughly 25 years.

Investor-level equity exposure, denoted $E_{i,t}$, is constructed as the investor's active equity adjustment scaled by lagged assets under management. Specifically, for each investor i and stock j, I compute the change in split-adjusted shares between quarters $t-1$ and t, $\Delta S_{i,j,t} = S_{i,j,t} - S_{i,j,t-1}$, and value this change using the stock's lagged price $P_{j,t-1}$. Aggregating across all stocks in the investor's portfolio and normalizing by lagged assets under management $AUM_{i,t-1}$ yields a measure of net equity holding change relative to portfolio size:

$$E_{i,t} = \frac{\sum_{j \in \mathcal{J}_{i,t}} P_{j,t-1} \, \Delta S_{i,j,t}}{AUM_{i,t-1}}. \tag{1}$$

This normalization ensures comparability across investors of different sizes and captures net buying or selling pressure. Positive values indicate net equity purchases, while negative values indicate net sales. To limit the influence of extreme adjustments, $E_{i,t}$ is winsorized at the 1st and 99th percentiles within each investor's time series. The final row in Table 1 reports summary statistics for equity exposure pooled across all investor-quarters. The mean is close to zero (0.01), indicating no systematic aggregate shift in equity exposure. However, the distribution is wide, with the 1st and 99th percentiles at -0.26 and 0.38, respectively, and a standard deviation of 0.13. These magnitudes indicate substantial dispersion in investor-level adjustments, suggesting that while aggregate changes are small, there is considerable heterogeneity in how individual investors adjust their equity exposure over time.

Table 4 reports summary statistics for the investor-level characteristics used in the heterogeneity analysis. The estimation sample consists of 2,118 investors, each with at least 60 quarterly observations, and an average time-series length of 97.5 quarters. Investor size is highly skewed: the mean average AUM is \$7,839 million, while the median is only \$640 million, reflecting the presence of a small number of very large institutions. This skewness is also evident in the wide dispersion of AUM, with an interquartile range from \$263 million to \$2,279 million and a standard deviation exceeding \$100 billion. The log transformation of AUM reduces this dispersion, with a mean of 6.75 and a standard deviation of 1.63. Trading intensity, measured by the average absolute active demand scaled by AUM, is relatively low on average (0.16) but exhibits substantial heterogeneity across investors, with a median of 0.062 and a long right tail.

Panel B reports the cross-sectional distribution of investor types. Investment advisors, which include hedge funds, constitute the majority of the sample, accounting for 69.5% of investors, followed by mutual funds (15.1%) and banks (9.6%), with smaller shares for insurance companies, pension funds and other institutional investors. Despite their numerical dominance, investment advisors account for only 23.7% of total average AUM, whereas mutual funds manage 33.8% and banks 10.3%. Notably, a single household investor represents

a negligible fraction of the sample in count but accounts for 26.7% of total AUM. Overall, these patterns indicate substantial heterogeneity in both the size and composition of institutional investors, which is central to the analysis of differential responses to aggregate risk conditions.

2.1 A Baseline One-Factor Expected-Return Specification

Before decomposing investor responses across individual risk-premia proxies, it is useful to ask a simpler question: do investors adjust aggregate equity exposure in response to variation in expected market returns at all? To answer this question, I begin with a one-factor specification that summarizes the information in dividend yield, the default premium, and the term spread into a single fitted expected-return measure.

Specifically, I first estimate predictive regressions of future aggregate market returns on the three business-cycle indicators introduced in Section 1. For a given horizon h, the market-level forecasting equation is

$$R^M_{t,t+h} = a_h + b_{DY,h} DY_t + b_{DEF,h} DEF_t + b_{TERM,h} TERM_t + u_{t+h}, \tag{2}$$

where $R^M_{t,t+h}$ denotes the cumulative market return from quarter t to horizon $t+h$, and DY_t, DEF_t, and $TERM_t$ denote dividend yield, the default premium, and the term spread, respectively. I then use the fitted value from this regression,

$$\widehat{ER}^{(h)}_t = \hat{a}_h + \hat{b}_{DY,h} DY_t + \hat{b}_{DEF,h} DEF_t + \hat{b}_{TERM,h} TERM_t, \tag{3}$$

as a summary measure of expected aggregate equity returns at horizon h.

In the second step, I estimate investor-specific sensitivities of equity exposure to this fitted expected-return measure:

$$E_{i,t} = \alpha^{(h)}_i + \gamma^{(h)}_i \widehat{ER}^{(h)}_t + \varepsilon_{i,t}, \tag{4}$$

where $E_{i,t}$ is investor i's active equity exposure scaled by lagged assets under management. The coefficient $\gamma_i^{(h)}$ captures how investor i adjusts equity exposure when the fitted expected market return rises. A positive coefficient indicates that the investor expands equity exposure when expected returns are high, consistent with absorbing aggregate risk when compensation is more attractive. A negative coefficient indicates that the investor reduces exposure in those states, consistent with shedding risk despite higher expected returns.

This one-factor specification provides a direct test of whether investors respond to expected returns *per se*. It is intentionally more parsimonious than the multi-proxy specification introduced below. Its advantage is interpretability: because the regressor is the fitted expected market return itself, the estimated coefficient maps directly into an investor's sensitivity to expected returns. Its limitation is that it imposes a common expected-return index across investors and therefore abstracts from the possibility that different institutions respond differently to different components of risk premia. I therefore use it as a baseline specification and then turn to the richer three-beta model to study heterogeneity across valuation, credit, and term-structure signals.

Table 3 reports results from a parsimonious one-factor expected-return specification. Panel A shows that the fitted expected-return measure captures meaningful variation in future market returns, particularly at longer horizons. The predictive power of the three-variable forecasting model increases substantially with horizon: the R^2 rises from 2.5% at the quarterly horizon to 7.0% at the annual horizon and 27.8% at the five-year horizon. The default premium emerges as the strongest predictor across horizons, while dividend yield and the term spread become increasingly informative at longer horizons. These results confirm that the fitted-value measure captures economically meaningful variation in expected returns and is particularly informative for medium- and long-horizon investment opportunities.

Panel B examines how institutions adjust aggregate equity exposure in response to this fitted expected-return measure. The central result is that only a minority of institutions exhibit a statistically significant response. Depending on the horizon used to construct

expected returns, between 16.5% and 18.1% of institutions have a significant fitted beta at the 5% level. Conversely, more than four-fifths of institutions display no measurable exposure adjustment with respect to the fitted expected-return series.

Among the institutions that do respond, positive responses are considerably more common than negative responses. Roughly 11% to 13% of institutions increase equity exposure when fitted expected returns rise, whereas only about 5% reduce exposure in the same states. Thus, significant responses are not only relatively uncommon, but they are also asymmetrically tilted toward institutions that expand rather than contract equity exposure when expected-return conditions improve.

The explanatory power of the fitted-value specification at the institution level is modest. Average investor-level R^2 values range from 2.6% to 2.8%, indicating that variation in expected returns explains only a small fraction of quarter-to-quarter changes in institutional equity exposure. This result is not surprising given the large number of idiosyncratic forces that influence institutional portfolio adjustments, including client flows, rebalancing needs, funding considerations, and institution-specific investment opportunities.

Overall, the fitted-value specification delivers two important insights. First, institutions differ substantially in how they respond to variation in expected returns, with only a minority exhibiting systematic exposure adjustments. Second, even among those institutions that respond, reactions are far from uniform. Some increase exposure when expected returns rise, while others move in the opposite direction. Because the fitted expected-return measure aggregates information from dividend yield, the default spread, and the term spread into a single index, however, it cannot reveal which underlying source of risk-premium variation drives these responses. The next section therefore allows each predictor to enter separately and examines whether institutions respond differently to valuation, credit, and term-structure conditions.

2.2 Type heterogeneity

To sharpen the interpretation of the first-stage results, I next examine which types of investors respond significantly to the fitted expected-return measure. For each horizon, I classify investors into three groups according to the estimated fitted-value beta from the one-factor specification: significantly positive, significantly negative, and insignificant. A significantly positive beta indicates that the investor increases equity exposure when the fitted expected market return rises, whereas a significantly negative beta indicates that the investor reduces exposure in those states. This classification provides a simple way to study not only whether investors respond to expected returns, but also which types of institutions are most likely to absorb or shed aggregate equity risk.

Table 5 shows that the dominant pattern is non-response. Across all three fitted-value specifications, the majority of investors in every type are statistically insignificant. In the full sample, 82.8% of investors are insignificant in the one-quarter specification, 81.9% in the one-year specification, and 83.5% in the five-year specification. Thus, the limited-participation result is not driven by a single investor class; it is pervasive across banks, insurers, advisors, mutual funds, pension funds, and other institutions.

Among the investors who do respond significantly, positive responses are more common than negative responses at every horizon. In the one-year specification, for example, 13.4% of all investors have significantly positive fitted-value betas, compared with only 4.7% with significantly negative betas. This asymmetry suggests that when investors do react to variation in expected returns, they are more likely to expand equity exposure when compensation rises than to cut exposure in the same states.

There is also meaningful heterogeneity across investor types. Measured by within-type response rates, pension funds and mutual funds are the most likely to respond positively to fitted expected returns. In the one-quarter specification, 23.5% of pension funds and 16.4% of mutual funds have significantly positive betas, compared with 11.5% of investment advisors and 9.7% of banks. The same pattern is visible at the one-year horizon, where

21.6% of pension funds and 19.7% of mutual funds respond positively. Investment advisors dominate the raw number of significant investors, but this is largely because they constitute the largest share of the sample. In propensity terms, pension funds and mutual funds appear more likely to absorb aggregate equity risk when expected returns rise.

Negative responses are less common and more evenly dispersed across types. Mutual funds and investment advisors account for much of the significantly negative group, with negative-response rates generally around 5% to 7%, while banks and pension funds exhibit relatively few significantly negative fitted-value betas. The single household observation is insignificant in all three specifications. Overall, these patterns suggest that fitted expected-return sensitivity is concentrated in a minority of institutions, with positive risk absorption most prevalent among pension funds and mutual funds.

2.3 Estimating investors' macro betas

The fitted-value specification provides a useful summary measure of expected returns, but it imposes a common expected-return index across all institutions. As a result, it cannot distinguish whether institutions respond differently to valuation conditions, credit conditions, or changes in the term structure. To examine these dimensions separately, I estimate institution-specific sensitivities to each of the three risk-premium proxies.

Specifically, I estimate the following time-series regression separately for each institution:

$$E_{i,t} = \alpha_i + \beta_i^{DY} DY_t + \beta_i^{DEF} DEF_t + \beta_i^{TERM} TERM_t + \varepsilon_{i,t}, \tag{5}$$

where $E_{i,t}$ denotes institution i's aggregate equity exposure, DY_t is dividend yield, DEF_t is the default spread, and $TERM_t$ is the term spread.

The coefficients β_i^{DY}, β_i^{DEF}, and β_i^{TERM} summarize how institution i's equity exposure varies with each of the three risk-premium proxies. A positive coefficient indicates that the institution tends to increase equity exposure when the corresponding variable rises, while a

negative coefficient indicates that the institution tends to reduce exposure in those states. The magnitude of the coefficient measures the strength of this relationship.

Estimating Equation 5 separately for each institution allows for substantial heterogeneity in exposure responses. Rather than imposing a representative-investor benchmark, this approach directly characterizes how institutions differ in their sensitivity to valuation conditions, credit conditions, and term-structure variation. These institution-specific coefficients, which I refer to as *macro betas*, form the basis of the subsequent analysis.

Before examining each risk-premium proxy separately, it is useful to summarize how common significant exposure responses are across institutions. To do so, I classify institutions according to the number of statistically significant macro betas estimated from Equation 5. An institution can have between zero and three significant coefficients, depending on whether its equity exposure is significantly related to dividend yield, the default spread, and the term spread.

Figure 7 presents the distribution. The most striking feature is the large fraction of institutions with no significant coefficient on any of the three indicators. Approximately 60% of institutions fall into this category. Another 28% have a significant loading on exactly one indicator, roughly 10% have significant loadings on two indicators, and only about 1% have significant loadings on all three. Thus, systematic exposure adjustments with respect to these commonly used measures of expected-return variation are concentrated in a minority of institutions.

Whether this concentration is economically large or small is not immediately clear. Institutional investors differ substantially in their mandates, constraints, investment horizons, and trading styles, and one would not necessarily expect all institutions to react to the same state variables. The evidence therefore does not imply that institutions ignore changes in expected returns, nor does it imply that institutions with insignificant coefficients are passive investors. Rather, it indicates that the three variables studied here explain exposure adjustments for some institutions much better than for others.

Table 7 examines the distribution of institution-specific coefficients in greater detail. The results reveal substantial dispersion in exposure sensitivities across institutions. Some institutions tend to increase equity exposure when a particular risk-premium proxy rises, others tend to reduce exposure, and many exhibit little measurable relation between exposure changes and the corresponding state variable. This heterogeneity is the central empirical pattern motivating the remainder of the paper.

Dividend yield. Dividend yield is one of the most widely studied predictors of stock returns and is commonly interpreted as a valuation-based measure of expected returns. In the sample used in this paper, a one-percentage-point increase in dividend yield predicts roughly a 3-percentage-point increase in cumulative returns over the next year and an approximately 11-percentage-point increase over the next five years.

Despite the predictive content of dividend yield, only 422 institutions, or approximately 20% of the sample, exhibit a statistically significant dividend-yield beta. Roughly 11% have positive coefficients, while 9% have negative coefficients. Thus, even for one of the most extensively studied predictors of expected returns, significant exposure responses are concentrated among a relatively limited subset of institutions.

The average dividend-yield beta is close to zero, but the cross-sectional dispersion is substantial. The standard deviation is 0.65, with estimates ranging from approximately -23 to $+8$. This wide distribution indicates that institutions differ markedly in how their equity exposure varies with valuation conditions. Some institutions tend to increase exposure when dividend yield rises, whereas others tend to reduce exposure, and the magnitude of these responses varies considerably across institutions.

Equally important, the majority of institutions display no detectable relationship between equity exposure and dividend yield. Approximately 80% of institutions have insignificant dividend-yield betas. Consequently, while dividend yield contains meaningful information about future market returns, its influence on institutional exposure adjustments is far from uniform. This pattern highlights the importance of examining which institutions adjust

exposure and what characteristics distinguish them from institutions whose exposure appears largely unrelated to valuation-based measures of expected returns.

The substantial dispersion in dividend-yield betas raises a natural question: which institutions tend to increase exposure when valuation-based expected-return measures rise, which institutions reduce exposure, and which display no systematic relationship at all? Understanding these differences is central to understanding how institutions respond to changing market conditions. The next section examines whether observable institutional characteristics help explain this variation.

Default premium. The default premium is among the strongest predictors of future market returns in the sample. As shown in Section 1, a one-percentage-point increase in the default premium predicts approximately a 1.5-percentage-point increase in next-quarter returns, a 4-percentage-point increase in next-year returns, and nearly a 20-percentage-point increase in returns over the subsequent five years.

Despite its predictive power, only 342 institutions, or approximately 16% of the sample, exhibit a statistically significant exposure response to changes in the default premium. Roughly 84% of institutions have insignificant default-spread betas, indicating that their aggregate equity exposure displays no detectable relation to this indicator over the sample period.

As with dividend yield, the average coefficient is close to zero but the cross-sectional dispersion is substantial. The standard deviation is approximately 0.70, with estimates ranging from -30 to $+4$. Some institutions tend to increase equity exposure when default spreads widen, while others tend to reduce exposure in the same states. The magnitude of these responses also varies considerably across institutions.

The combination of strong predictive power at the market level and limited exposure sensitivity at the institutional level is noteworthy. Although the default premium contains significant information about future market returns, only a relatively small subset of institutions exhibits exposure adjustments that are systematically related to this signal. Con-

sequently, institutions differ not only in the direction of their response to credit conditions, but also in whether they respond at all.

Term spread. The term spread is another widely used indicator of business-cycle conditions and expected returns. In the sample studied here, its predictive content is concentrated primarily at longer horizons, where it captures persistent variation in economic conditions and expected market returns.

A total of 315 institutions, or approximately 15% of the sample, display a statistically significant term-spread beta. Among these institutions, positive sensitivities are substantially more common than negative sensitivities. Approximately 12% of institutions increase equity exposure when the term spread rises, whereas only about 3% exhibit the opposite pattern.

The mean coefficient is again close to zero, but the cross-sectional distribution is wide, indicating considerable variation in how institutions' equity exposure relates to changes in the term structure. Thus, as with dividend yield and the default premium, institutions differ substantially in both the sign and magnitude of their exposure sensitivities.

Across all three specifications, the explanatory power of the regressions is modest on average but varies substantially across institutions. The mean investor-level R^2 is approximately 8%, with a standard deviation of 7%. For some institutions, dividend yield, the default premium, and the term spread explain virtually none of the variation in equity exposure. For others, these variables account for as much as 60% of exposure variation. Among institutions with at least one statistically significant macro beta, the average R^2 increases to roughly 13%.

Taken together, these results reveal substantial heterogeneity in institutions' exposure sensitivities to common measures of market risk premia. Some institutions display positive sensitivities, others negative sensitivities, and many exhibit no statistically detectable relationship between equity exposure and the three state variables considered here. Moreover, the fraction of institutions with significant coefficients differs across indicators, suggesting that valuation conditions, credit conditions, and term-structure conditions affect institutional

exposure adjustments in different ways.

The remainder of the paper examines whether this heterogeneity is systematically related to observable institutional characteristics. In particular, I investigate whether investor type, investment horizon, size, and age help explain why institutions differ in their sensitivity to dividend yield, default spreads, and term spreads.

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Table 7 about here

3 What Drives Heterogeneity?

The previous section documented substantial heterogeneity in institutions' exposure sensitivities to market risk-premium proxies. Some institutions tend to increase equity exposure when risk-premium measures rise, others tend to reduce exposure, and many exhibit little measurable relationship between exposure adjustments and these state variables. The objective of this section is to develop an economic framework for interpreting these differences.

Three broad mechanisms may generate heterogeneity in estimated macro betas. First, institutions may differ in their funding capacity and balance-sheet constraints. Second, they may differ in investment horizon and therefore place different weights on slow-moving variation in expected returns. Third, institutions may interpret the same information differently, leading to heterogeneous portfolio responses even in the absence of differences in constraints or horizons. These mechanisms are not mutually exclusive. Rather, they provide complementary explanations for why institutions adjust exposure differently when market conditions change.

The goal of this section is not to estimate a structural model of institutional demand. Instead, it provides a conceptual framework for interpreting the cross-sectional variation in macro betas and motivates the investor characteristics examined later in the paper.

3.1 Funding Capacity and Institutional Constraints

A natural explanation for differences in institutions' exposure responses is variation in funding capacity. A large literature emphasizes that institutions' ability to adjust risk exposures depends not only on expected returns but also on balance-sheet strength, access to funding, and regulatory constraints. When funding conditions deteriorate, institutions may be unable or unwilling to expand risky positions even when expected returns become more attractive.

This mechanism is central to intermediary asset-pricing models such as Brunnermeier and Pedersen (2009), He and Krishnamurthy (2013b), and Gromb and Vayanos (2018). In these

models, institutions with stronger balance sheets can adjust exposures more aggressively when investment opportunities improve, whereas constrained institutions may display weaker responses or may even move exposure in the opposite direction.

Funding capacity is difficult to observe directly for many institutional investors. For a subset of banks and insurance companies, regulatory filings provide information on capitalization, leverage, and regulatory buffers. For the broader set of institutions, direct measures of balance-sheet strength are often unavailable. I therefore rely on observable characteristics that may proxy for funding stability and the ability to reallocate capital. Institution size, fund flows, and other measures of financial flexibility may capture some of these differences. The central prediction is that institutions with greater funding capacity should exhibit stronger exposure responses to changes in expected returns than otherwise similar institutions facing tighter constraints.

3.2 Investment Horizon

A second mechanism is investment horizon. Many commonly used predictors of expected returns, including dividend yield and the term spread, contain information about future returns primarily at medium and long horizons. Institutions differ substantially in the horizons over which portfolio performance is evaluated, creating the possibility that the same expected-return signal may be highly relevant for some investors but much less relevant for others.

Even in the absence of binding constraints, institutions with short evaluation horizons may place relatively little weight on slow-moving variation in expected returns. Benchmark considerations, performance evaluation, and client pressures may lead such institutions to focus on short-run outcomes rather than long-run investment opportunities. Institutions with longer horizons, by contrast, may be better positioned to respond to persistent changes in expected returns because they are less sensitive to short-term fluctuations in performance.

This mechanism appears frequently in the institutional-investor literature. Long-horizon

institutions can potentially react more strongly to valuation and business-cycle signals, whereas short-horizon institutions may exhibit weaker sensitivities or no measurable response at all. Consequently, investment horizon provides a natural explanation for why otherwise similar institutions may display very different macro betas.

To capture differences in horizon, I construct three related measures. First, I calculate portfolio turnover based on gross trading activity scaled by lagged assets under management. Higher turnover indicates more frequent portfolio reallocation and is commonly interpreted as reflecting shorter investment horizons.

Second, I measure active-demand intensity using the average absolute magnitude of active equity-demand adjustments scaled by lagged assets under management. This measure captures how aggressively an institution changes aggregate equity exposure over time and therefore reflects trading intensity rather than conventional turnover.

Third, I construct a holding-spell-based horizon measure. For each institution, I calculate the average duration of uninterrupted stock-level holding spells. Unlike turnover measures, holding spells directly capture how long positions remain in a portfolio. Together, these variables provide complementary information about how actively and how persistently institutions manage their equity positions.

3.3 Beliefs and Interpretation of Market Conditions

Differences in funding capacity and investment horizon may not fully explain heterogeneity in exposure responses. Even among institutions with similar resources and similar horizons, portfolio adjustments may differ because institutions interpret the same information differently.

This idea is a central feature of models with heterogeneous beliefs. Changes in dividend yield, default spreads, or the term spread may have multiple interpretations. One institution may view a rise in dividend yield as evidence that equities have become relatively attractive, while another may interpret the same signal as indicating weaker future economic prospects.

Likewise, widening default spreads may be viewed either as an investment opportunity or as a sign of increasing macroeconomic risk.

These differences in interpretation can generate variation in exposure responses even when institutions face similar constraints and have comparable investment horizons. Although beliefs are difficult to observe directly, they provide a natural explanation for why institutions sometimes respond in opposite directions to the same market conditions.

The empirical analysis in this paper does not attempt to directly measure investor beliefs. Instead, beliefs should be viewed as a residual channel that may help explain exposure variation that remains after accounting for observable differences in funding capacity, investment horizon, and institutional characteristics.

Taken together, funding capacity, investment horizon, and beliefs provide a useful framework for understanding heterogeneity in institutions' exposure sensitivities to market risk-premium proxies. Motivated by these mechanisms, the next section examines whether observable characteristics such as investor type, size, age, trading activity, and investment horizon help explain the cross-sectional variation in estimated macro betas.

3.4 Investor-Level Variables

To examine whether observable investor characteristics help explain the heterogeneity in macro betas documented earlier, I construct four investor-level variables: age, size, turnover, and investor type. These variables proxy for differences in experience, risk-bearing capacity, investment horizon, and institutional mandates. This subsection describes how each variable is defined and motivates its economic relevance.

Age. Investor age is defined as the number of quarters an institution appears in the 13F filings. For investor i, let t_i^{first} denote the first quarter in which the investor reports holdings and t_i^{last} the last quarter in the sample. Age is computed as

$$\text{Age}_i = t_i^{\text{last}} - t_i^{\text{first}} + 1.$$

Age captures institutional maturity, reporting history, and organizational stability. Older institutions may have more established investment processes, longer planning horizons, and more stable client bases, while younger institutions may face greater performance pressure or shorter evaluation cycles. Although age is an imperfect proxy for experience or discipline, it provides a simple and transparent measure of institutional longevity.

Size. Investor size is measured using assets under management (AUM). For each investor i, I compute the time-series average of quarterly AUM:

$$AUM_i = \frac{1}{T_i} \sum_{t=1}^{T_i} AUM_{i,t},$$

where $AUM_{i,t}$ is the total market value of the investor's equity holdings in quarter t. I use $\log(AUM_i)$ in the empirical analysis to reduce the influence of extreme values and to interpret size effects more naturally. Size is theoretically relevant because larger institutions tend to have greater risk-bearing capacity, more stable funding, and more diversified portfolios. In intermediary asset-pricing models, well-capitalized intermediaries behave more pro-cyclically and expand risk exposure when expected returns rise, whereas smaller institutions face tighter constraints and adjust more cautiously.

Turnover. Turnover captures how frequently an investor reshuffles its equity portfolio. Following the standard holdings-based measure, I compute investor-quarter turnover as one-half of the absolute value of purchases plus sales during the quarter, scaled by lagged assets under management. Using holdings data, purchases and sales are inferred from changes in split-adjusted shares valued at lagged prices. Specifically, for investor i in quarter t,

$$\text{Turnover}_{i,t} = \frac{1}{2} \frac{\sum_j P_{j,t-1} \left| \Delta S_{i,j,t} \right|}{AUM_{i,t-1}}, \qquad \Delta S_{i,j,t} = S_{i,j,t} - S_{i,j,t-1}.$$

I then define investor i's overall turnover as the time-series average of quarterly turnover:

$$\text{Turnover}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} \text{Turnover}_{i,t}.$$

This measure is standard in the literature as a proxy for trading intensity and, indirectly, investment horizon. Investors with low turnover adjust their portfolios less frequently and are typically interpreted as longer-horizon investors, whereas investors with high turnover trade more actively and are typically interpreted as shorter-horizon investors.

Investor Type. Each investor is assigned to one of the standard institutional categories used in the asset-pricing literature, following the Thomson Reuters/WRDS classification: investment advisors, mutual funds, banks, insurance companies, pension funds, households, and a residual "other" category. Investor type captures broad differences in regulation, mandate, liability structure, and balance-sheet flexibility. For example, pension funds and insurance companies typically operate under long-dated liabilities and stable funding, while investment advisors and mutual funds face shorter evaluation horizons and greater flow-driven trading pressure. These institutional differences provide a natural source of heterogeneity in how investors respond to business-cycle conditions and, therefore, in how much equity risk they absorb.

Together, these four variables—age, size, turnover, and investor type—provide a set of observable characteristics that proxy for the theoretical mechanisms discussed in Section 3. The next section presents summary statistics for these variables and examines their explanatory power for the cross-sectional variation in macro betas.

3.5 Descriptive statistics

Table 4 reports summary statistics for the four characteristics. Average AUM is highly skewed: the mean is roughly \$8 billion, while the median is only \$0.6 billion, reflecting the presence of many small institutions alongside a few very large ones. Trading intensity is

low on average but varies widely, indicating substantial differences in how actively investors rebalance their portfolios. The average investor appears in the data for 97 quarters (about 24 years), with a wide interquartile range.

Panel B of the table shows the distribution of investor types. Investment advisors constitute 70% of the sample but collectively hold only 24% of total AUM. Mutual funds represent 15% of investors but hold 34% of AUM. Banks account for 10% of investors and 10% of AUM, while insurance companies and pension funds each represent small shares of both counts and assets. The household sector is represented by a single large filer that holds 27% of total AUM. These patterns highlight the coexistence of many small institutions and a handful of very large asset managers.

3.6 Second-stage regression

To explain the cross-sectional heterogeneity in macro sensitivity, I estimate the following second-stage regression:

$$\beta_{k,i} = \alpha_k + \gamma_{1,k} \log(AUM_i) + \gamma_{2,k} Turnover_i + \gamma_{3,k} Age_i + \delta_k' Type_i + u_{k,i}, \tag{6}$$

for each risk premia indicator $k \in \{DY, DEF, TERM\}$, where $\hat{\beta}_{k,i}$ is investor i's estimated first-stage sensitivity of equity exposure to risk premian indicator k.

This specification asks whether differences in investor size, trading intensity, age, and institutional type help explain sign and intensity of investors responsiveness to expected return indicators. The regression is estimated separately for the dividend-yield beta, the default-premium beta, and the term-spread beta, allowing the determinants of macro sensitivity to differ across predictors.

The interpretation of the coefficients is straightforward. A positive $\gamma_{1,k}$ implies that larger investors exhibit a more positive exposure response to macro variable k. A positive $\gamma_{2,k}$ implies that more active traders respond more aggressively to changes in that macro

condition. A positive $\gamma_{3,k}$ implies that older institutions—those with longer reporting histories—respond more strongly. The coefficients on the type indicators show whether a given investor class is more or less sensitive than the omitted benchmark category, holding size, turnover, and age fixed.

Because the dependent variable in this stage is itself estimated, the second-stage regression should be interpreted as descriptive rather than causal. Its purpose is to identify which observable investor attributes are systematically associated with stronger or weaker macro sensitivity in equity exposure.

Table 8 reports results for both the full sample of investors and the subsample of investors whose first-stage betas are statistically significant. The restricted sample helps isolate true behavioral differences from statistical noise, since roughly 80 percent of investors have betas close to zero. The results are reported for fitted value betas in the first columns and then in thenxts, for 3 factors.

Table 8 about here

The fitted-value specification shows little systematic heterogeneity when all investors are included, but much sharper differences once attention is restricted to the subset of investors with statistically significant fitted-value betas. In the full sample of 2,087 investors, the regression explains only a small share of the cross-sectional variation in fitted-value exposure, with an R^2 of 0.04, and none of the investor characteristics is statistically significant. This suggests that, for the typical investor, exposure to the fitted expected-return factor is weak and not tightly linked to observable investor characteristics.

The picture changes substantially in the significant-only sample. Among the 359 investors whose fitted-value beta is significant at the 5% level, the explanatory power rises sharply, with an R^2 of 0.72. The strongest pattern is turnover: investors with higher turnover exhibit significantly lower fitted-value betas, while investors with lower turnover load more positively on the fitted expected-return factor. This is consistent with the idea that the investors most

responsive to expected returns are relatively less aggressive traders. Size also matters in this subsample: larger investors, measured by log average AUM, have more positive fitted-value betas. By contrast, investor age does not appear to explain fitted-value exposure. Across investor types, most coefficients are not statistically different from the omitted category of investment advisors, although pension funds stand out with significantly more negative fitted-value betas in the significant-only sample. Overall, the fitted-value results suggest that heterogeneity is concentrated in a relatively small subset of investors, and within that group, trading intensity and size are more informative than broad institutional type.

The following columns report the results for the three business-cycle variables separately. I discuss each variable in detail below.

Size (log AUM). Investor size plays only a limited role. It is statistically significant only for dividend-yield betas in the restricted sample, with a coefficient of 0.04. This implies that larger investors tend to increase their equity exposure when dividend yield rises. Since a higher dividend yield is associated with higher expected returns or higher discount rates, this pattern suggests that larger institutions behave more contrarian: they raise exposure when valuations appear attractive. One interpretation is that large investors face fewer trading frictions, have more stable funding, or possess longer horizons, enabling them to lean into valuation signals rather than away from them.

Turnover. Turnover is the most robust and economically meaningful characteristic. It is strongly negative for all three macro variables in the restricted sample, and negative for dividend yield even in the full sample. A negative coefficient means that high-turnover investors—those who actively and frequently adjust their portfolios—tend to reduce equity exposure when risk premia rise. For example, a one-unit increase in turnover is associated with a 1.53-unit lower dividend-yield beta and a 0.64-unit lower term-spread beta. This suggests that high-turnover investors behave more pro-cyclically or risk-averse: when macro indicators point to higher risk premia or weaker conditions, they cut exposure more aggres-

sively. These investors may be more sensitive to short-term performance, face tighter risk controls, or operate under mandates that require rapid de-risking.

Age. Investor age, measured as the number of quarters an investor appears in the data, has little explanatory power. The only marginally significant effect is a small negative coefficient for default-premium betas in the restricted sample. Overall, longevity in the dataset does not systematically shape macro sensitivity.

Investor type. Investor type matters primarily for the term spread. Banks and pension funds have significantly lower term-spread betas than hedge funds in the restricted sample. This is intuitive: banks and pensions are typically more constrained, more liability-driven, and less flexible in adjusting equity exposure in response to macroeconomic signals. Hedge funds, by contrast, are more opportunistic and more likely to increase exposure when the yield curve steepens and economic conditions improve. For dividend yield and default premium, type differences are small and statistically weak.

Overall, turnover emerges as the dominant driver of macro sensitivity, followed by investor type for the term spread. Size plays a modest role, and age plays almost none. These results suggest that the cross-section of macro sensitivity is shaped primarily by differences in trading aggressiveness and institutional constraints rather than by simple scale or tenure.

Table 8 about here

The cross-sectional patterns in macro sensitivity align closely with theoretical predictions about how institutional frictions, investment horizons, and portfolio mandates shape investor behavior. In standard asset-pricing models, time-varying expected returns imply that investors should adjust their risk exposure when business-cycle indicators move. However, models with heterogeneous agents—such as those featuring trading frictions, delegated portfolio management, or liability-driven constraints—predict that not all investors will respond in the same way.

The strong role of turnover is consistent with theories in which more active or short-horizon investors react more aggressively to changes in risk premia. High-turnover institutions may face tighter risk controls, more frequent performance evaluation, or greater sensitivity to short-term volatility, leading them to reduce exposure when macro indicators signal rising risk premia. Conversely, low-turnover investors behave more like long-horizon allocators, adjusting exposure less or even leaning into valuation signals.

The modest effect of size fits theories of scale economies and reduced frictions: larger investors may have better access to liquidity, lower trading costs, or more stable funding, enabling them to increase exposure when expected returns rise. This is consistent with contrarian behavior in models where well-capitalized investors absorb risk when valuations become attractive.

The importance of investor type for term-spread sensitivity echoes theoretical distinctions between constrained and unconstrained institutions. Banks and pension funds, which operate under regulatory capital requirements or liability-matching mandates, adjust exposure less when the yield curve steepens. Hedge funds, by contrast, resemble the flexible, risk-seeking agents in heterogeneous-agent models who expand risk-taking when economic conditions improve.

Overall, the empirical results support a view of the equity market in which macroeconomic shocks are intermediated by a diverse set of institutions with different frictions, horizons, and constraints. Business-cycle variation in expected returns does not translate uniformly into portfolio adjustments; instead, it generates a rich cross-section of responses that reflect the underlying economic roles and limitations of different investor types.

3.7 Investor Classification by Risk-Premia Sensitivity

To complement the regression-based analysis, I classify investors according to the sign and statistical significance of their macro betas. For each macro variable, investors fall into one of three groups: (i) significantly positive beta, (ii) significantly negative beta, or (iii)

insignificant beta. This classification provides a direct way to identify which institutions tend to absorb aggregate risk when expected returns rise and which tend to shed it.

Across all three macro variables, the majority of investors fall into the insignificant category, consistent with the first-stage results. However, the subset of investors with significant betas displays systematic differences in size, turnover, age, and institutional type. Panels A–C of Table 8 summarize these patterns.

3.7.1 Dividend Yield

Dividend-yield sensitivities display a balanced but economically meaningful split: 11% of investors have significantly positive betas, while 9% have significantly negative betas. Size differences across groups are minimal. In contrast, turnover varies sharply. Investors with negative betas exhibit more than twice the turnover of those with positive betas (0.18 vs. 0.08), indicating substantially shorter investment horizons. Age differences are modest, with positive-beta investors being slightly older on average (107 vs. 100 quarters). Overall, positive-beta investors appear to be longer-horizon, lower-turnover institutions.

Panel C reveals strong differences across investor types. Mutual funds, insurance companies, and pension funds are disproportionately represented among positive-beta investors. For example, 23% of mutual funds have positive betas compared with 10% with negative betas, and 17% of insurance companies have positive betas compared with only 1% negative. Pension funds show a similar pattern (21% vs. 8%). These patterns suggest that long-horizon institutions play a central role in absorbing risk associated with dividend-yield variation. In contrast, investment advisors and banks display more balanced distributions, while households show no systematic response.

3.7.2 Default Spread

Default-spread sensitivities are more symmetric: 9% of investors have significantly positive betas and 8% have significantly negative betas. Size differences remain small, but turnover

again distinguishes the groups. Negative-beta investors have nearly double the turnover of positive-beta investors (0.15 vs. 0.08), reinforcing the distinction between short- and long-horizon institutions. Age differences are negligible.

Across investor types, pension funds are more likely to have positive betas (21% vs. 8%), consistent with their long-horizon risk-absorbing role. Mutual funds and insurance companies tilt slightly toward negative betas, while investment advisors show little asymmetry. These results suggest that institutions differ in how they adjust equity exposure to changes in credit-related risk premia.

3.7.3 Term Spread

Term-spread sensitivities exhibit the strongest asymmetry. Twelve percent of investors have significantly positive betas, while only 3% have significantly negative betas. Turnover differences are particularly pronounced: negative-beta investors have extremely high turnover (0.29 vs. 0.08), indicating very short horizons. Positive-beta investors are slightly older on average.

Investor-type patterns differ from the other two variables. Investment advisors and banks are more likely to have positive term-spread betas, suggesting that these institutions adjust their equity exposure more strongly in response to changes in the term structure. This pattern is consistent with the interpretation of the term spread as capturing variation in growth expectations and duration-related risk premia.

Taken together, the sign-based classification reinforces the regression evidence. Long-horizon, low-turnover institutions—such as pension funds, insurance companies, and some mutual funds—tend to increase equity exposure when risk premia rise. In contrast, high-turnover, short-horizon investors tend to reduce exposure, behaving more pro-cyclically. These patterns highlight the central role of investment horizon and trading aggressiveness in shaping how institutions respond to business-cycle conditions.

Table 9 about here

3.7.4 Does Investor Type Matter?

To further examine heterogeneity across institutional groups, I estimate type-level aggregate regressions in which equity exposure is aggregated within each investor type and regressed on the three business-cycle variables. This exercise complements the investor-level analysis by asking whether entire institutional categories exhibit systematic sensitivity to aggregate risk premia.

Across the six institutional groups, statistically significant macro sensitivities are concentrated in three types: banks, pension funds, and mutual funds.

Banks. Banks display a negative and statistically significant sensitivity to dividend yield and a positive and significant sensitivity to the term spread. Quantitatively, a one-percentage-point increase in dividend yield is associated with a 0.0028 decline in banks' aggregate equity exposure, while a one-percentage-point increase in the term spread is associated with a 0.0027 increase. Thus, banks reduce exposure when valuation-based expected returns rise, but increase exposure when the yield curve steepens. This pattern is consistent with banks' sensitivity to funding conditions and regulatory constraints, which may limit their ability to lean into valuation signals but allow them to expand risk-taking when economic conditions improve.

Pension funds. Pension funds move in the opposite direction with respect to dividend yield. A one-percentage-point increase in dividend yield is associated with a 0.0034 increase in pension funds' aggregate equity exposure. This suggests that pension funds behave more like long-horizon risk absorbers, increasing exposure when expected returns are high. Their liability-driven investment mandates and long-term horizons may allow them to take advantage of valuation opportunities that shorter-horizon institutions avoid.

Mutual funds. Mutual funds respond significantly to the default premium. A one-percentage-point increase in the default spread is associated with a 0.0048 decline in mutual

funds' aggregate equity exposure. This behavior is consistent with mutual funds reducing risk when macro-financial conditions deteriorate and required compensation for risk rises. Performance evaluation pressures and flow-driven constraints may amplify this pro-cyclical response.

Other types. The remaining type-level coefficients are statistically insignificant. Investment advisors, insurance companies, and households do not exhibit systematic aggregate sensitivity to any of the three business-cycle variables once exposures are aggregated within type.

Overall, the type-level regressions reinforce the investor-level findings: heterogeneity in macro sensitivity is not random but systematically related to institutional form. Banks, pension funds, and mutual funds respond to different dimensions of aggregate risk premia, and the direction of adjustment varies across types. These results highlight that institutional roles, constraints, and horizons shape how different investor groups intermediate business-cycle fluctuations in expected returns.

Table 10 about here

4 A Simple Model of Heterogeneous Exposure Responses

The empirical analysis documents two central facts. First, institutions differ substantially in the extent to which their equity exposure responds to changes in market risk-premium proxies. While a minority of institutions exhibit statistically significant exposure sensitivities, many display little measurable relationship between exposure adjustments and these state variables. Second, among those institutions that do respond, both the sign and magnitude of estimated sensitivities vary considerably. Long-horizon, low-turnover institutions are more likely to increase exposure when risk-premium proxies rise, whereas high-turnover institutions are more likely to reduce exposure in the same states.

This section develops a simple framework that rationalizes these patterns. The model incorporates two ingredients emphasized in the literature on institutional investors and intermediary asset pricing: (i) time-varying expected returns driven by a persistent state variable, and (ii) heterogeneous institutions facing different horizon- and constraint-induced tradeoffs. The objective is not to provide a fully specified equilibrium model, but rather to illustrate how differences in horizons and constraints can generate heterogeneous exposure responses to common changes in expected returns.

4.1 Environment

Time is discrete. A single risky asset delivers return

$$R_{t+1} = r_f + \lambda x_t + \varepsilon_{t+1},$$

where r_f is the risk-free rate, x_t is a persistent state variable capturing variation in expected returns, and ε_{t+1} is an i.i.d. innovation with variance σ^2. $The state variable evolves according to x_{t+1} = \rho x_t + \eta_{t+1}, \qquad \rho \in (0,1).$

The variable x_t summarizes the common component of expected-return variation captured by dividend yield, the default spread, and the term spread. A higher value of x_t

corresponds to a state in which expected equity returns are higher.

There are two institution types indexed by $j \in \{L, S\}$. Long-horizon institutions, denoted by L, evaluate portfolio performance over a relatively long horizon. Short-horizon institutions, denoted by S, face stronger interim-performance concerns, benchmark pressures, or funding-related constraints. Each institution chooses a portfolio share θ_t^j invested in the risky asset.

4.2 Long-Horizon Institutions

Long-horizon institutions maximize expected utility of terminal wealth:

$$\max_{\{\theta_t^L\}} \mathbb{E}_t \left[U(W_T^L) \right].$$

Under a standard mean-variance approximation, the optimal risky share is

$$\theta_t^L = \frac{\mathbb{E}_t[R_{t+1} - r_f]}{\gamma_L \mathrm{Var}_t(R_{t+1})} = \frac{\lambda x_t}{\gamma_L \sigma^2},$$

where γ_L denotes risk aversion.

Differentiating with respect to the state variable yields

$$\frac{\partial \theta_t^L}{\partial x_t} = \frac{\lambda}{\gamma_L \sigma^2} > 0.$$

Long-horizon institutions therefore increase equity exposure when expected returns rise. Because they place greater weight on long-run investment opportunities, they respond positively to persistent variation in expected returns.

4.3 Short-Horizon Institutions

Short-horizon institutions face additional costs associated with expanding risky positions. These costs may reflect flow pressure, benchmark concerns, funding constraints, regulatory

requirements, or other frictions that become more severe in states where risk premia are elevated.

Their objective function is

$$\max_{\theta_t^S} \ \mathbb{E}_t[W_{t+1}^S] - \frac{\gamma_S}{2}\mathrm{Var}_t(W_{t+1}^S) - \frac{\phi}{2}\kappa(x_t)\theta_t^{S\,2},$$

where $\gamma_S > \gamma_L$ captures greater sensitivity to short-run risk, $\phi > 0$ measures the importance of interim constraints, and $\kappa(x_t)$ is increasing in x_t.

The function $\kappa(x_t)$ captures the idea that periods associated with high expected returns may simultaneously involve tighter funding conditions, greater uncertainty, or stronger benchmark concerns.

Solving yields

$$\theta_t^S = \frac{\lambda x_t}{\gamma_S \sigma^2 + 2\phi\kappa(x_t)}.$$

Differentiating with respect to x_t gives

$$\frac{\partial \theta_t^S}{\partial x_t} = \frac{\lambda\left(\gamma_S\sigma^2 + 2\phi\kappa(x_t)\right) - \lambda x_t\left(2\phi\kappa'(x_t)\right)}{\left(\gamma_S\sigma^2 + 2\phi\kappa(x_t)\right)^2}.$$

If constraints become sufficiently important when expected returns rise,

$$2\phi x_t \kappa'(x_t) > \gamma_S\sigma^2 + 2\phi\kappa(x_t),$$

then

$$\frac{\partial \theta_t^S}{\partial x_t} < 0.$$

In this case, short-horizon institutions optimally reduce equity exposure despite the increase in expected returns. The marginal cost of expanding risky positions rises more rapidly than the increase in expected compensation.

4.4 Implications and Testable Predictions

The model generates three predictions that map directly to the empirical analysis.

Prediction 1 (Sign Heterogeneity). Long-horizon institutions exhibit positive exposure sensitivities:

$$\beta^L > 0.$$

Short-horizon or constrained institutions may exhibit weaker sensitivities or even negative sensitivities:

$$\beta^S \leq 0.$$

Consequently, the cross-section of institutions should display substantial variation in both the magnitude and sign of estimated macro betas.

Prediction 2 (Role of Investment Horizon). Institutions with longer horizons should respond more strongly to persistent variation in expected returns. Empirically, institutions with lower turnover and longer holding periods should exhibit more positive exposure sensitivities than institutions with high turnover and short holding periods.

Prediction 3 (Limited Reactivity). Only institutions with sufficiently low constraint intensity ϕ exhibit economically meaningful exposure responses to changes in x_t. As constraint intensity increases, exposure adjustments become smaller and eventually indistinguishable from zero. Consequently, only a subset of institutions will display statistically significant macro betas, while many institutions will appear largely insensitive to common measures of expected-return variation.

The model formalizes a simple intuition: institutions differ in how they translate changes in expected returns into portfolio adjustments. Long-horizon institutions place greater weight on persistent variation in expected returns and therefore tend to increase exposure when expected-return opportunities improve. Short-horizon or constrained institutions face coun-

tervailing forces that weaken or even reverse this response. These differences generate substantial heterogeneity in exposure sensitivities and provide a simple interpretation for the wide dispersion in estimated macro betas documented in the data.

5 Conclusion

This paper studies how U.S. institutional investors adjust aggregate equity exposure in response to variation in expected returns over the business cycle. Using investor-level holdings data, I estimate institution-specific sensitivities of equity exposure to three widely used predictors of expected returns: dividend yield, the default spread, and the term spread. Together, these measures provide a tractable characterization of time-varying market risk premia and allow me to examine how institutions adjust their portfolios as expected-return conditions evolve.

Three empirical findings emerge. First, systematic exposure responses are concentrated in a minority of institutions. For most institutions, changes in equity exposure display little measurable relationship to dividend yield, the default spread, or the term spread. Second, among institutions that do respond, exposure sensitivities are highly heterogeneous. Some institutions tend to increase exposure when risk-premium proxies rise, while others tend to reduce exposure in the same states. Third, this heterogeneity follows a systematic pattern. Long-horizon, low-turnover institutions are more likely to exhibit positive exposure sensitivities, whereas high-turnover institutions are more likely to exhibit weaker or negative sensitivities. Responses also differ across institutional types, with pension funds, insurance companies, banks, and investment advisors displaying distinct sensitivity profiles across valuation, credit, and term-structure conditions.

These findings suggest that changes in expected returns do not translate into uniform portfolio adjustments across institutional investors. Instead, institutions differ substantially in how their aggregate equity exposure responds to common measures of market conditions. Investment horizon appears particularly important in explaining these differences, while investor type provides additional explanatory power. In contrast, observable characteristics such as age and size account for relatively little of the variation in estimated macro betas.

More broadly, the results highlight the importance of institutional heterogeneity in understanding the transmission of macroeconomic conditions into investor demand. Asset-pricing

theories often emphasize variation in expected returns, yet much less is known about how different institutions react to those changes. By documenting large differences in exposure sensitivities across institutions, this paper provides new evidence on the demand side of asset pricing and identifies a set of institutional characteristics that help explain why investors respond differently to the same expected-return signals.

Several questions remain for future research. In particular, a deeper understanding of the roles of funding capacity, investment horizon, and heterogeneous beliefs may help explain why some institutions respond strongly to changes in expected returns while others exhibit little measurable sensitivity. Understanding these mechanisms more fully may provide additional insights into how institutional demand shapes asset prices over the business cycle.

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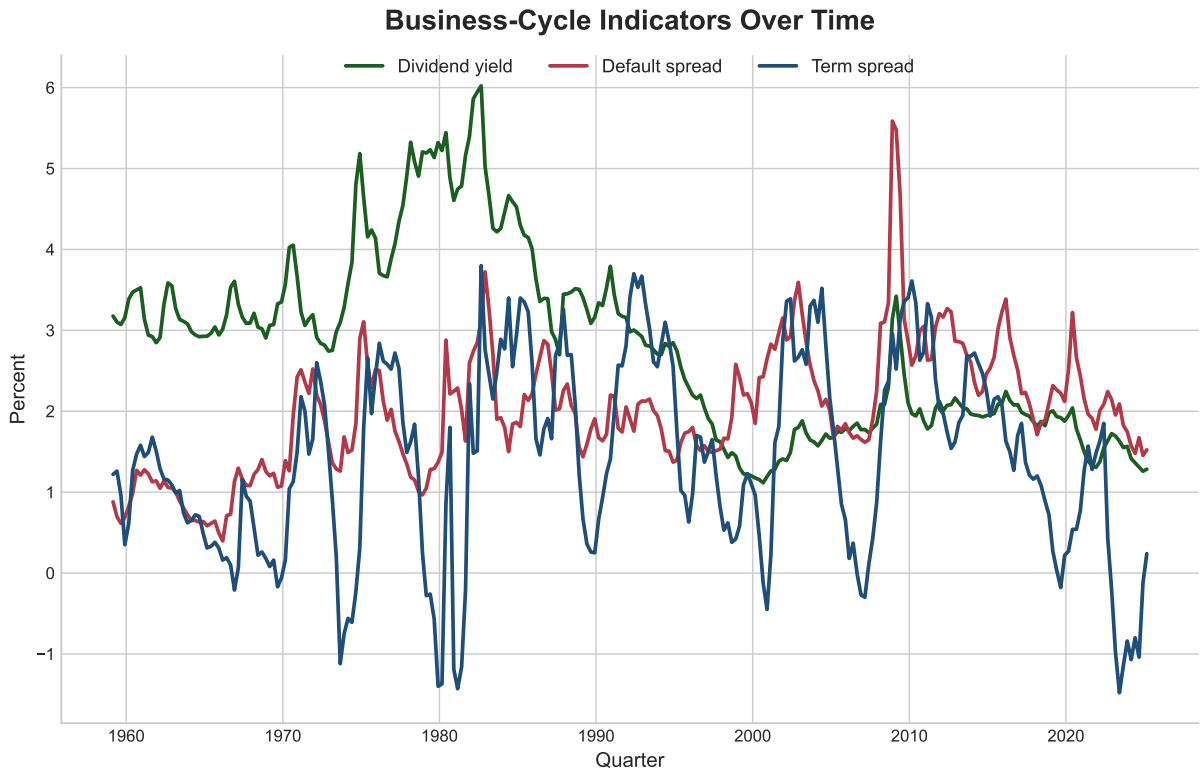
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Notes: Dividend yield is the S&P 500 dividend yield. Default spread is the BAA-AAA corporate bond spread. Term spread is the 10-year Treasury yield minus the 3-month Treasury bill rate. Sample: 1959Q1--2025Q1.

Figure 1: This figure plots the time series of three aggregate risk-premia indicators: dividend yield, default premium, and term spread. Dividend yield is the quarterly S&P 500 dividend yield and captures valuation-based variation in expected equity returns. Default premium is the spread between BAA-rated corporate bond yields and long-term Treasury yields, and proxies for variation in compensation for bearing credit and macroeconomic risk. Term spread is defined as the difference between the 10-year Treasury yield and the 3-month Treasury bill rate, and captures variation in the slope of the yield curve and expected future economic conditions. All series are quarterly and obtained from FRED-QD.

6 Appendix

This section shows the figures and tables used in the body of the paper.

6.1 Figures

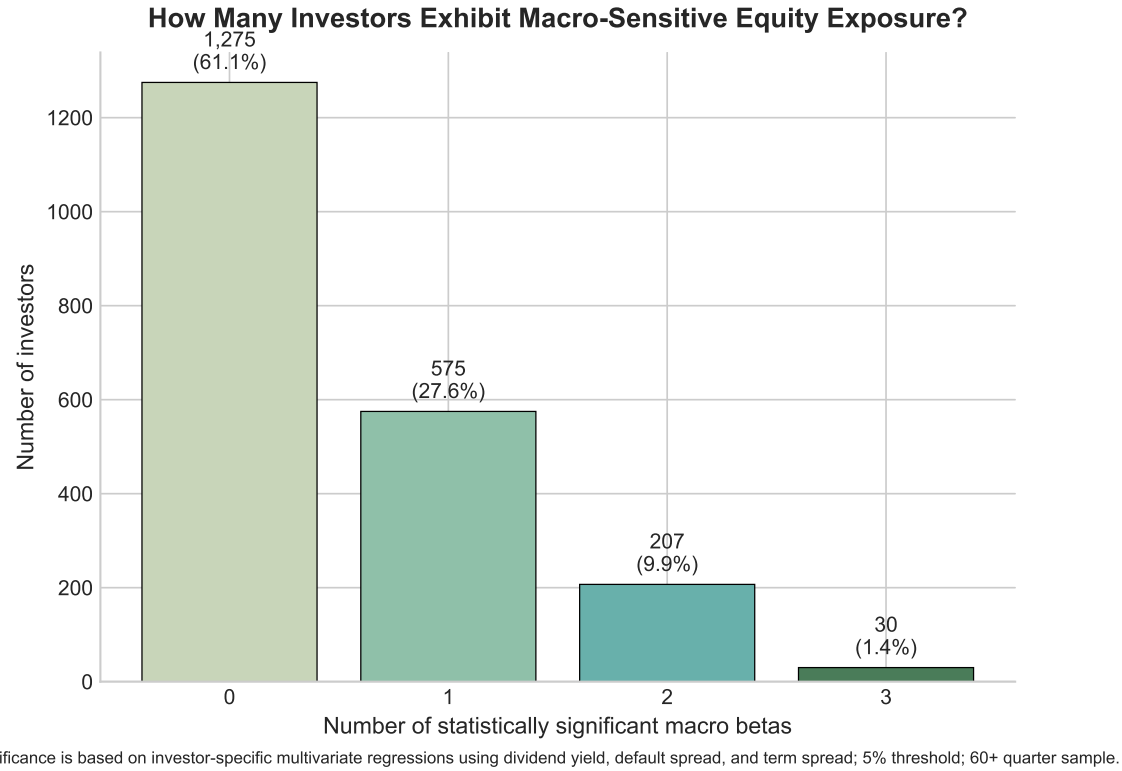


Figure 2: Business-cycle sensitivity across investors. The figure plots the distribution of investors by the number of statistically significant macro sensitivities in the first-stage multivariate regression of investor-level equity exposure on dividend yield, default spread, and term spread. Each bar reports the number of investors with 0, 1, 2, or 3 significant macro betas at the 5 percent level. The sample is restricted to investors with at least 60 quarterly observations.

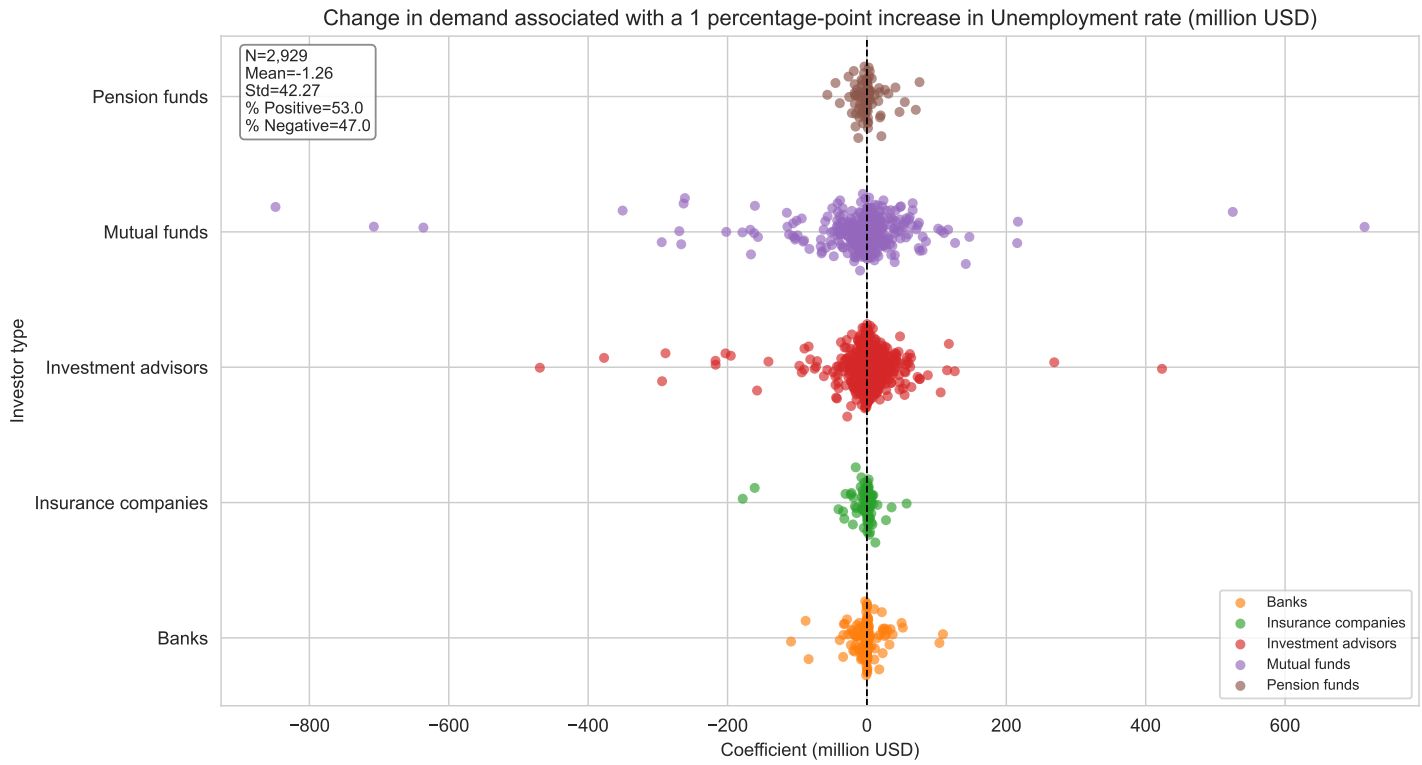


Figure 3: Caption

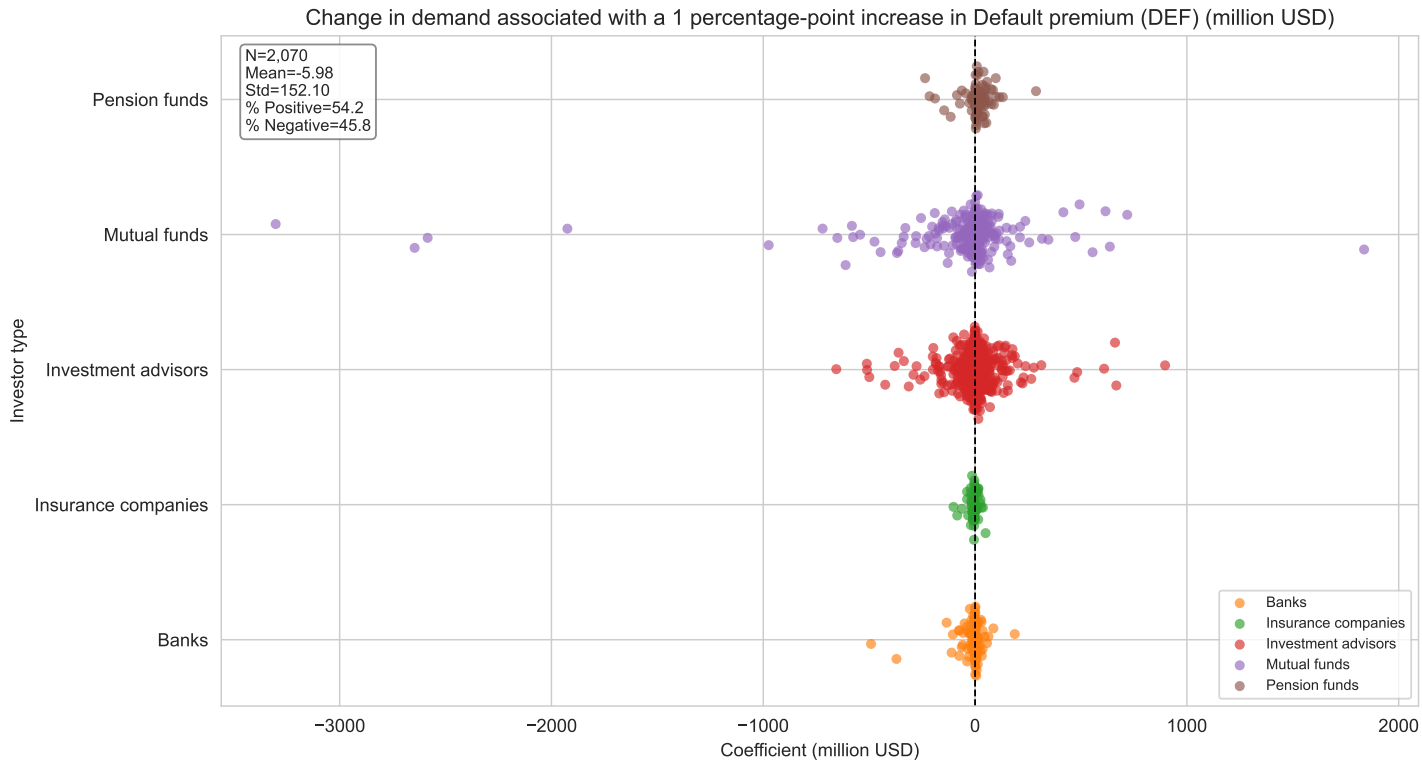


Figure 4: Caption

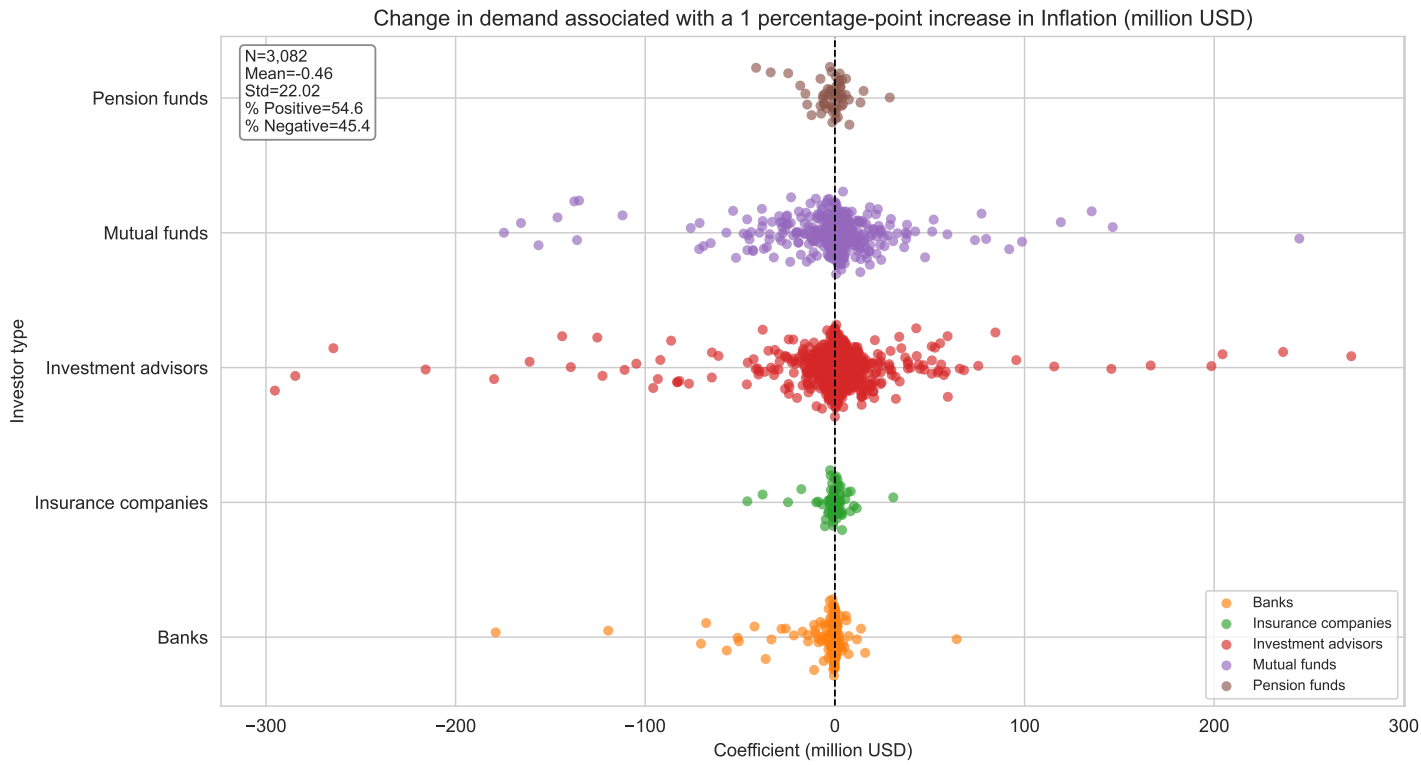


Figure 5: Caption

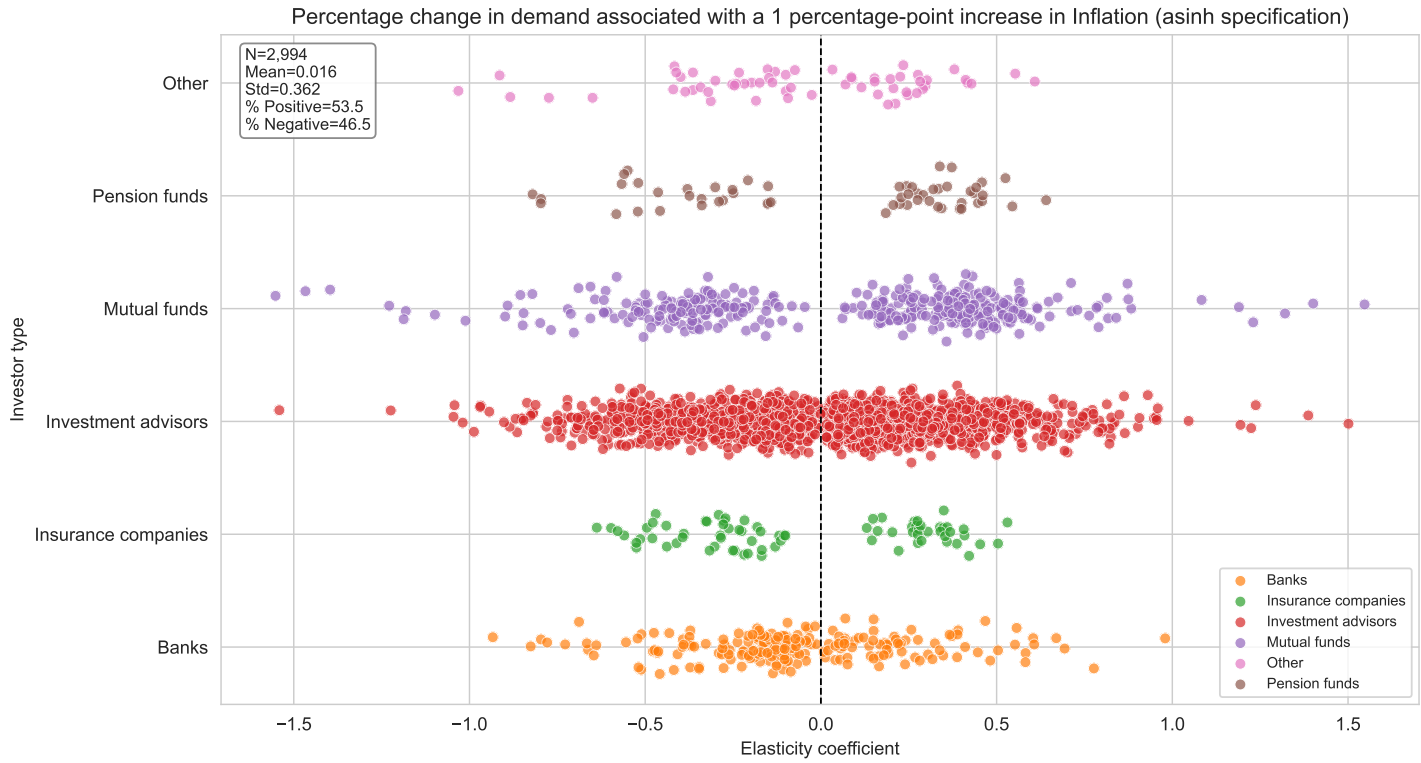


Figure 6: Caption

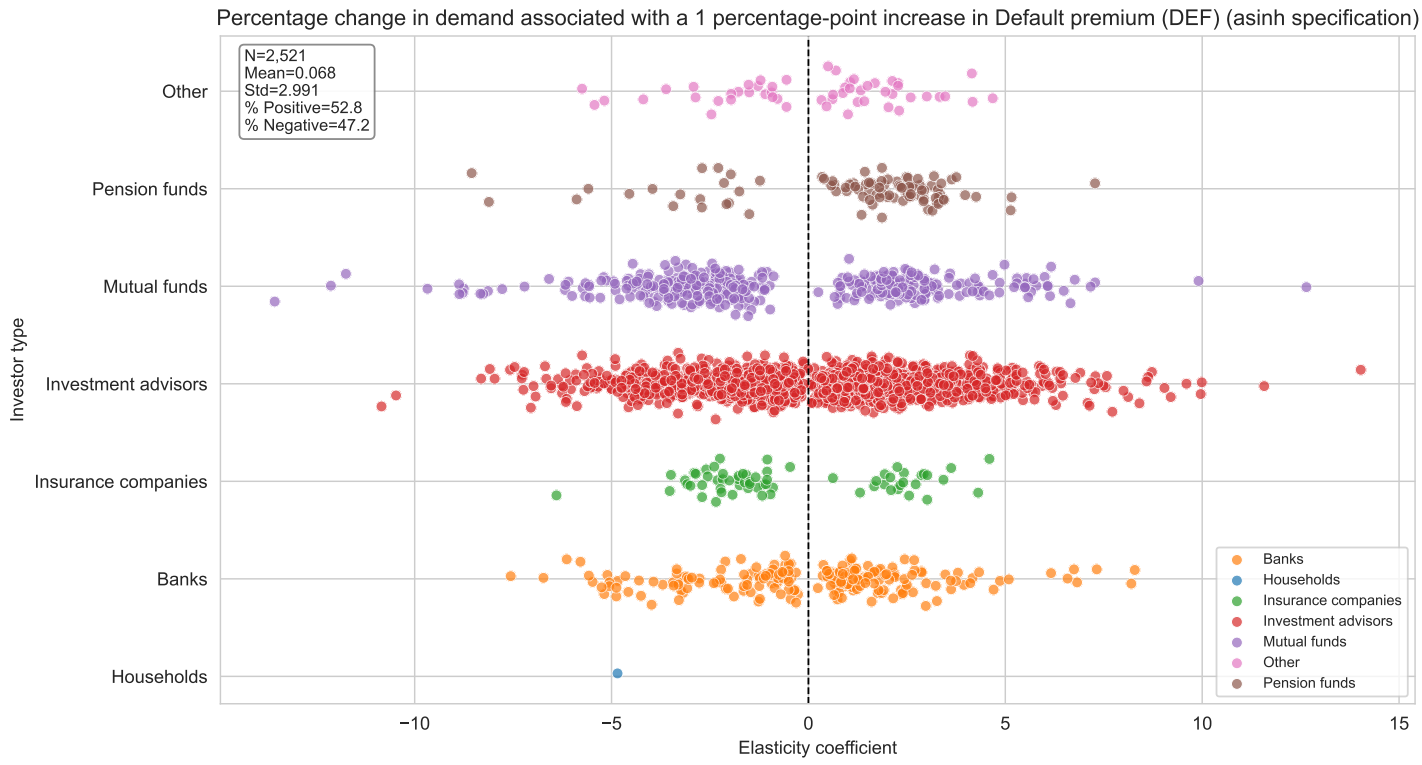


Figure 7: Caption

Graph 7 shows

6.2 Tables

Table 1: Summary Statistics and Correlations for Business-Cycle Variables, Market Return, and Investor Demand

Variable	1st pct.	Mean	99th pct.	SD
Panel A. Summary statistics				
Dividend yield	1.17	2.48	5.88	1.11
Default spread	1.43	2.30	4.86	0.67
Term spread	-1.25	1.65	3.68	1.25
Market return	-20.78	2.63	19.96	8.43
Equity Exposure Changes	-0.26	0.01	0.38	0.13
Panel B. Correlations				
	Dividend yield	Default spread	Term spread	Market return
Dividend yield	1.00	0.06	0.23	0.01
Default spread	0.06	1.00	0.31	-0.07
Term spread	0.23	0.31	1.00	-0.02
Market return	0.01	-0.07	-0.02	1.00

This table reports summary statistics and correlations for the main business-cycle variables, aggregate market return, and investor demand used in the analysis. Dividend yield is the quarterly S&P 500 dividend yield from FRED-QD (`S&P div yield`). Default spread is the BAA–AAA corporate bond yield spread from FRED-QD (`BAA10YM`). Term spread is the slope of the Treasury yield curve, measured as the 10-year Treasury yield minus the 3-month Treasury bill rate from FRED-QD (`GS10TB3Mx`). Market return is the quarterly Fama–French aggregate market return constructed as the quarterly excess market return plus the risk-free rate, $R_{m,t} = (MKT - RF)_t + RF_t$. Demand is the quarterly cross-sectional mean of investor-level equity adjustment scaled by lagged assets under management, `ad_over_aum_lag`, where investor-level active equity adjustment is aggregated from holdings data and normalized by lagged investor AUM. Panel A reports the 1st percentile, mean, 99th percentile, and standard deviation. Panel B reports pairwise correlations among the three business-cycle variables and market return. The merged quarterly sample spans 1980Q2–2024Q3.

Table 2: Predictability of Aggregate Market Returns by Business-Cycle Variables

	Univariate			Multivariate
	DY	DEF	TERM	(All)
Panel A. 1-quarter ahead returns				
Dividend yield	0.59			0.74
	(1.21)			(1.48)
Default spread		1.35*		1.49*
		(1.68)		(1.66)
Term spread			0.46	0.04
			(1.00)	(0.08)
Observations	246	246	246	246
R^2	0.01	0.01	0.00	0.02
Panel B. 1-year ahead returns				
Dividend yield	2.24			2.71*
	(1.46)			(1.83)
Default spread		4.19**		4.28*
		(2.13)		(1.87)
Term spread			1.97	0.82
			(1.41)	(0.49)
Observations	243	243	243	243
R^2	0.02	0.03	0.02	0.07
Panel C. 5-year ahead returns				
Dividend yield	7.80			10.91**
	(1.35)			(2.31)
Default spread		24.02***		19.58**
		(3.90)		(2.29)
Term spread			15.06***	9.96
			(2.70)	(1.54)
Observations	227	227	227	227
R^2	0.04	0.16	0.15	0.28

This table reports predictive regressions of cumulative aggregate market returns on standard business-cycle variables. The dependent variable is the cumulative market return over the next h quarters, constructed by compounding quarterly returns. Panel A uses $h = 1$, Panel B $h = 4$, and Panel C $h = 20$ quarters. Dividend yield (DY) is the S&P 500 dividend yield, default spread (DEF) is the BAA–AAA corporate bond yield spread, and term spread (TERM) is the 10-year Treasury yield minus the 3-month Treasury bill rate. Univariate specifications include each predictor separately, while the multivariate specification includes all three jointly. Entries are OLS coefficients with Newey–West t-statistics in parentheses. HAC lag length is $h-1$ to account for overlapping returns. The sample spans 1963Q3–2025Q2. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: Fitted Expected Returns and Investor-Level Equity-Exposure Responses

	One-quarter ahead	One-year ahead	Five-year ahead
Panel A. Market Return Forecasting Regressions			
R^2	0.025	0.070	0.278
Dividend yield	0.741	2.711***	10.906***
Default premium	1.490*	4.280***	19.583***
Term spread	0.041	0.822	9.959***
N	246	243	227
Panel B. Investor-Level Fitted-Beta Regressions			
Share significant at 5%	0.172	0.181	0.165
Share significantly positive	0.121	0.134	0.114
Share significantly negative	0.051	0.047	0.051
Mean investor-level R^2	0.026	0.027	0.028
Median investor-level R^2	0.009	0.009	0.011
Mean fitted beta ($\gamma_i^{(h)}$)	-0.0249	-0.0036	-0.0002
Median fitted beta ($\gamma_i^{(h)}$)	0.0015	0.0006	0.0001
Std. dev. of fitted betas	0.7335	0.1676	0.0301
Investors	2,087	2,087	2,087

This table reports results from a baseline one-factor expected-return specification. Panel A reports predictive regressions of future cumulative market returns on dividend yield, the default spread, and the term spread. The reported coefficients are slope estimates from Equation 2. Panel B reports institution-level regressions of quarterly equity exposure on the fitted expected-return series obtained from Panel A, as in Equation 4. The investor sample consists of all institutions with at least 60 quarters of holdings history. Significantly positive and negative shares are based on two-sided tests at the 5% significance level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 4: Investor Characteristics: Estimation Sample

Variable	*N*	Mean	SD	P25	Median	P75	
Panel A. Continuous investor characteristics							
Size	2,118	7,839.762	103,937.347	262.693	639.777	2,279.262	
Log (Size)	2,118	6.750	1.627	5.571	6.461	7.732	
Turnover	2,118	0.160	1.043	0.040	0.062	0.104	
Age	2,118	97.519	30.925	73.000	90.000	115.000	
Panel B. Investor type distribution							
Type	Count	Percent					Size Share
Investment advisors	1,471	69.45					23.70
Mutual funds	320	15.11					33.83
Banks	204	9.63					10.25
Insurance companies	73	3.45					2.92
Pension funds	49	2.31					2.64
Households	1	0.05					26.66

This table reports summary statistics for the investor-level characteristics used in the heterogeneity analysis. The sample is restricted to investors with at least 60 quarterly observations, yielding 2,118 distinct investors. Average AUM is the time-series average of investor-level assets under management, constructed from quarterly equity holdings. Log average AUM is the natural logarithm of average AUM. Average absolute active demand / AUM is the time-series mean of the absolute value of investor-level active equity adjustment scaled by lagged assets under management, and serves as a measure of trading intensity. Number of quarters is the total number of quarterly observations available for each investor in the estimation sample. Panel B reports the cross-sectional distribution of investor types. The final column reports each type's share of the total investor-level average AUM in the estimation sample, computed as the sum of average AUM across investors within a type divided by the sum of average AUM across all investors.

Table 5: Investor Responses to Fitted Expected Returns by Investor Type

Horizon	Investor type	Total	Insignificant	Significant +	Significant −
Panel A. One-quarter-ahead fitted expected return					
	Overall	2,087	1,728 (82.8%)	253 (12.1%)	106 (5.1%)
	Banks	226	199 (88.1%)	22 (9.7%)	5 (2.2%)
	Insurance companies	72	61 (84.7%)	8 (11.1%)	3 (4.2%)
	Investment advisors	1,370	1,143 (83.4%)	157 (11.5%)	70 (5.1%)
	Mutual funds	305	234 (76.7%)	50 (16.4%)	21 (6.9%)
	Pension funds	51	36 (70.6%)	12 (23.5%)	3 (5.9%)
	Other	62	54 (87.1%)	4 (6.5%)	4 (6.5%)
	Households	1	1 (100.0%)	0 (0.0%)	0 (0.0%)
Panel B. One-year-ahead fitted expected return					
	Overall	2,087	1,709 (81.9%)	279 (13.4%)	99 (4.7%)
	Banks	226	199 (88.1%)	22 (9.7%)	5 (2.2%)
	Insurance companies	72	62 (86.1%)	8 (11.1%)	2 (2.8%)
	Investment advisors	1,370	1,126 (82.2%)	173 (12.6%)	71 (5.2%)
	Mutual funds	305	228 (74.8%)	60 (19.7%)	17 (5.6%)
	Pension funds	51	38 (74.5%)	11 (21.6%)	2 (3.9%)
	Other	62	55 (88.7%)	5 (8.1%)	2 (3.2%)
	Households	1	1 (100.0%)	0 (0.0%)	0 (0.0%)
Panel C. Five-year-ahead fitted expected return					
	Overall	2,087	1,743 (83.5%)	238 (11.4%)	106 (5.1%)
	Banks	226	198 (87.6%)	23 (10.2%)	5 (2.2%)
	Insurance companies	72	65 (90.3%)	4 (5.6%)	3 (4.2%)
	Investment advisors	1,370	1,145 (83.6%)	150 (10.9%)	75 (5.5%)
	Mutual funds	305	234 (76.7%)	51 (16.7%)	20 (6.6%)
	Pension funds	51	44 (86.3%)	6 (11.8%)	1 (2.0%)
	Other	62	56 (90.3%)	4 (6.5%)	2 (3.2%)
	Households	1	1 (100.0%)	0 (0.0%)	0 (0.0%)

This table classifies investors by the sign and statistical significance of their fitted-value beta from the one-factor expected-return specification. For each horizon, the fitted expected return is obtained from the market-level forecasting regression of future market returns on dividend yield, the default premium, and the term spread. An investor is classified as *Significant +* if the fitted-value beta is positive with a two-sided $p < 0.05$, as *Significant −* if the fitted-value beta is negative with a two-sided $p < 0.05$, and as *Insignificant* otherwise. Percentages are reported within investor type.

Table 6: Elastic Investors by Type and Macroeconomic Variable

Type	Statistic	GDP	Unemp.	Infl.	Term	Default	Wealth	Elastic in type
Inv. advisors	Number	175	407	293	457	280	339	1,263
	Percent (%)	7.09	16.48	11.87	18.51	11.34	13.73	51.15
	Mean β	0.07	0.15	0.03	0.35	0.21	-0.04	–
	Mean R^2	–	–	–	–	–	–	0.22
Pension funds	Number	5	12	9	14	11	12	35
	Percent (%)	7.25	17.39	13.04	20.29	15.94	17.39	50.72
	Mean β	-0.14	0.37	0.21	0.22	2.29	-0.09	–
	Mean R^2	–	–	–	–	–	–	0.20
Banks	Number	22	45	23	50	27	34	126
	Percent (%)	7.41	15.15	7.74	16.84	9.09	11.45	42.42
	Mean β	0.25	0.18	-0.08	0.45	0.63	-0.04	–
	Mean R^2	–	–	–	–	–	–	0.18
Mutual funds	Number	35	107	46	70	71	55	237
	Percent (%)	9.19	28.08	12.07	18.37	18.64	14.44	62.20
	Mean β	0.20	0.31	0.17	0.34	-0.62	-0.10	–
	Mean R^2	–	–	–	–	–	–	0.16
Insurance	Number	5	13	13	18	9	13	44
	Percent (%)	6.10	15.85	15.85	21.95	10.98	15.85	53.66
	Mean β	-0.02	0.44	-0.30	0.20	-1.90	-0.13	–
	Mean R^2	–	–	–	–	–	–	0.16
Other	Number	13	15	11	11	13	13	47
	Percent (%)	11.82	13.64	10.00	10.00	11.82	11.82	42.73
	Mean β	0.30	0.06	-0.14	0.46	0.38	-0.04	–
	Mean R^2	–	–	–	–	–	–	0.22
Household	Number	0	1	0	1	1	0	1
	Percent (%)	0	100	0	100	100	0	100
	Mean β	–	2.35	–	-1.84	-5.77	–	–
	Mean R^2	–	–	–	–	–	–	0.07

Types are ordered by average R^2 among elastic investors (last column).
Percent (%) is relative to total investors within each institution type.
Mean β is computed over significant coefficients ($p < 0.05$).
Mean R^2 is computed for investors in that type with at least one significant macro coefficient.

Table 7: Investor-Level Equity Exposure and Business-Cycle Variables: Distribution of Regression Estimates

	Significance		Distribution of Estimates			
Parameter	N	(%)	Mean	SD	Min	Max
Dividend yield	422	20.22	-0.02	0.65	-23.09	8.37
Default premium	342	16.39	-0.01	0.70	-30.42	4.29
Term spread	315	15.09	0.00	0.17	-6.35	3.11
Alpha	435	20.84	0.07	2.20	-9.37	75.04
R^2 (all investors)	–	–	0.08	0.07	0.00	0.67
R^2 (at least one significant beta)	–	–	0.13	0.08	0.02	0.67
Sample Information						
Investors			2,087			
Investors with at least one significant beta			812			
Mean quarters per investor			97.08			

This table summarizes investor-specific multivariate regressions of equity exposure on business-cycle variables. For each investor with at least 60 quarterly observations, I estimate

$$E_{i,t} = \alpha_i + \beta_i^{DY} DY_t + \beta_i^{DEF} DEF_t + \beta_i^{TERM} TERM_t + \varepsilon_{i,t}.$$

The dependent variable, $E_{i,t}$, is investor-level active equity adjustment scaled by lagged assets under management, expressed in percentage terms. Dividend yield is the quarterly S&P 500 dividend yield from FRED-QD. Default premium is the BAA–AAA corporate bond yield spread. Term spread is the 10-year Treasury yield minus the 3-month Treasury bill rate. The dependent variable is winsorized at the 1st and 99th percentiles separately within each investor; the macro predictors are not winsorized. For each parameter, the table reports the number and percentage of investors with a two-sided p-value below 0.05, together with the mean, standard deviation, minimum, and maximum of the estimated coefficient across investors. The first R^2 row reports the distribution of fit across all investor-specific regressions, while the second reports the distribution restricted to investors with at least one statistically significant macro beta. The merged sample spans 1980Q3–2024Q3 and contains 2,087 investors.

Table 8: Investor Characteristics and Heterogeneity in Macro Sensitivity

	Fitted-value beta		Dividend-yield beta		Default-premium beta		Term-spread beta	
	All	Sig. only	All	Sig. only	All	Sig. only	All	Sig. only
SIZE (log AUM)	-0.00	0.04**	0.00	0.04*	-0.01	0.01	0.00	0.01
	(-0.42)	(2.04)	(0.06)	(1.78)	(-1.12)	(1.29)	(0.89)	(0.98)
Turnover	-0.12	-2.05***	-0.20**	-1.53**	-0.08	-2.99***	-0.02	-0.64***
	(-0.43)	(-9.57)	(-2.32)	(-2.05)	(-0.48)	(-37.71)	(-0.67)	(-2.93)
Age (quarters)	0.00	0.00	0.00	0.00	0.00	-0.00*	0.00	0.00
	(0.30)	(0.06)	(1.34)	(0.33)	(1.03)	(-1.87)	(0.58)	(0.78)
Investor type, omitted category: investment advisors								
Banks	0.01	0.15	-0.03	-0.06	-0.01	-0.01	-0.01	-0.06***
	(0.35)	(0.75)	(-1.45)	(-1.46)	(-0.47)	(-0.13)	(-1.34)	(-2.86)
Insurance companies	-0.39	-0.43	-0.24	-0.85	-0.00	0.03	-0.10	-0.32
	(-1.37)	(-1.59)	(-0.79)	(-1.15)	(-0.04)	(0.42)	(-1.12)	(-1.51)
Mutual funds	-0.06	-0.09	-0.03	0.04	-0.11	-0.04	-0.01	-0.02
	(-0.96)	(-1.51)	(-0.87)	(1.19)	(-1.22)	(-0.96)	(-0.55)	(-0.66)
Pension funds	0.01	-0.23**	-0.04	-0.12	0.01	-0.04	-0.01	-0.07***
	(0.25)	(-2.57)	(-1.08)	(-1.29)	(0.94)	(-0.67)	(-1.34)	(-3.83)
Households	-0.00	0.00	-0.10	0.00	0.05	0.00	-0.03	0.00
	(-0.00)	(.)	(-0.86)	(-1.16)	(0.97)	(0.99)	(-0.93)	(0.61)
Constant	0.02	-0.02	-0.07	-0.20	0.05	0.29***	-0.01	-0.03
	(0.24)	(-0.11)	(-0.79)	(-0.92)	(1.04)	(2.96)	(-0.52)	(-0.42)
Observations	2,087	359	2,087	422	2,087	342	2,087	315
R^2	0.04	0.92	0.11	0.62	0.02	0.96	0.02	0.73

This table reports second-stage cross-sectional regressions relating investor-specific macro sensitivities to observable investor characteristics. The first pair of columns uses the investor-specific coefficient from the first-stage regression of equity exposure on the fitted one-quarter expected-return factor. The remaining column pairs use investor-specific coefficients from first-stage multivariate time-series regressions of equity exposure on dividend yield, the default premium, and the term spread.

The "All" columns include all investors with at least 60 quarterly observations. The "Sig. only" columns restrict the sample, separately for each first-stage coefficient, to investors whose corresponding first-stage beta is statistically significant at the 5% level. Explanatory variables include investor-type indicators, log average assets under management, average turnover, and investor age measured as the number of quarters observed. Investment advisors are the omitted category. Entries report OLS coefficients with heteroskedasticity-robust t-statistics in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9: Sign and Significance of Investor-Level Macro Betas

	Dividend Yield			Default Premium			Term Spread		
	+	−	Insig.	+	−	Insig.	+	−	Insig.
Panel A. Distribution									
Number of investors	238	184	1,665	182	160	1,745	258	57	1,772
Percent of sample	11.40	8.82	79.78	8.72	7.67	83.61	12.36	2.73	84.91
Panel B. Investor characteristics									
Log average AUM	7.14	6.91	6.70	6.85	7.11	6.73	7.04	6.59	6.73
Average abs. demand / AUM	0.08	0.18	0.17	0.08	0.15	0.17	0.08	0.29	0.17
Age (quarters)	107.37	99.73	96.57	100.88	100.25	97.59	99.48	95.86	97.94
Panel C. Share within investor type (%)									
Investment advisors	9.08	9.22	81.70	8.45	7.28	84.27	13.17	2.63	84.20
Banks	5.91	7.88	86.21	7.88	0.99	91.13	6.90	1.48	91.63
Mutual funds	22.81	9.38	67.81	9.06	12.81	78.13	15.00	2.81	82.19
Insurance companies	16.67	1.39	81.94	6.94	11.11	81.94	4.17	5.56	90.28
Pension funds	20.83	8.33	70.83	20.83	8.33	70.83	6.25	6.25	87.50
Households	0.00	0.00	100.00	0.00	0.00	100.00	0.00	0.00	100.00

Table 10: Type-Level Aggregate Equity Exposure and Business-Cycle Variables

	Banks	Households	Insurance cos.	Inv. advisors	Mutual funds	Pension funds
Dividend yield	-0.002758**	-0.000541	-0.001969	-0.001510	0.000455	0.003389**
	(-2.27)	(-0.37)	(-1.22)	(-1.00)	(0.29)	(2.13)
Default premium	-0.000445	0.000457	-0.004843	0.003111	-0.004762**	0.004137
	(-0.19)	(0.17)	(-1.37)	(1.43)	(-2.47)	(1.14)
Term spread	0.002700**	0.000274	-0.001487	-0.000437	0.001218	-0.000025
	(2.44)	(0.16)	(-0.63)	(-0.34)	(1.13)	(-0.02)
Constant	0.006521	0.002301	0.020005	0.004211	0.014017	-0.020927
Observations	177	177	177	177	177	177
R^2	0.0291	0.0006	0.0244	0.0251	0.0323	0.0694

This table reports time-series regressions of type-level aggregate equity exposure on the three business-cycle variables. For each investor type and quarter, aggregate active demand is constructed by summing investor-level active demand across all investors in that type, and aggregate assets under management are constructed by summing investor-level AUM across the same investors. Type-level equity exposure is then defined as aggregate active demand divided by lagged aggregate AUM. To reduce the effect of outliers, the dependent variable is winsorized within type at the 1st and 99th percentiles. Each column reports a separate regression of the form

$$E_{g,t} = \alpha_g + \beta_{1,g} DY_t + \beta_{2,g} DEF_t + \beta_{3,g} TERM_t + \varepsilon_{g,t},$$

where $E_{g,t}$ is aggregate equity exposure for investor type g in quarter t, DY_t is the aggregate dividend yield, DEF_t is the default premium, and $TERM_t$ is the term spread. The sample is restricted to the 60+-quarter investor sample used in the baseline analysis, and spans 1980Q3–2024Q3. Entries are OLS coefficients with Newey–West t-statistics in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11: Future Portfolio Returns by Macro-Cyclicality Group

	Positive beta	Negative beta	Insignificant beta
Panel A. Classification by dividend-yield beta			
Investors	238	184	1,665
Next-quarter return (%)	3.24	2.99	2.98
Next-year cumulative return (%)	13.37	12.34	12.37
Next-5-year cumulative return (%)	81.70	69.64	73.45
Next-5-year annualized return (%)	11.53	10.11	10.49
Panel B. Classification by default-premium beta			
Investors	182	160	1,745
Next-quarter return (%)	3.04	3.09	3.00
Next-year cumulative return (%)	12.65	12.88	12.43
Next-5-year cumulative return (%)	75.29	76.64	73.69
Next-5-year annualized return (%)	10.81	10.70	10.54
Panel C. Classification by term-spread beta			
Investors	258	57	1,772
Next-quarter return (%)	3.06	3.14	3.00
Next-year cumulative return (%)	12.74	12.80	12.43
Next-5-year cumulative return (%)	74.60	80.75	73.76
Next-5-year annualized return (%)	10.78	11.28	10.53

This table reports future raw portfolio returns for investors classified by the sign and significance of their first-stage multivariate macro betas. For each investor with at least 60 quarterly observations, I estimate a time-series regression of investor-level equity exposure on dividend yield, default premium, and term spread. For each macro variable, investors are assigned to one of three groups: significantly positive beta, significantly negative beta, or statistically insignificant beta, using a 5 percent significance threshold. Investor-level raw portfolio returns are constructed from quarterly holdings using lagged portfolio weights and subsequent stock-level CRSP quarterly total returns. Delisting-adjusted returns are used only when ordinary CRSP returns are missing. One-year and five-year returns are obtained by compounding quarterly investor-level returns over four and twenty quarters, respectively. Reported returns are equal-weighted across investors, where each investor contributes its own average future return over the available sample. The sample is restricted to investors with at least 60 quarterly observations.

Table 12: Risk-Adjusted Performance by Macro-Cyclicality Group

	Positive beta	Negative beta	Insignificant beta
Panel A. Classification by dividend-yield beta			
Investors	238	184	1,665
CAPM alpha (annualized, %)	3.21	2.98	2.89
	(4.78)	(4.18)	(4.74)
Market beta	1.02	1.04	1.01
FF5 alpha (annualized, %)	2.67	2.47	2.36
	(5.23)	(5.28)	(5.34)
Sharpe ratio (annualized)	0.73	0.71	0.71
Panel B. Classification by default-premium beta			
Investors	182	160	1,745
CAPM alpha (annualized, %)	3.13	2.90	2.95
	(4.66)	(3.92)	(4.80)
Market beta	1.03	1.05	1.01
FF5 alpha (annualized, %)	2.69	2.56	2.40
	(6.08)	(4.64)	(5.36)
Sharpe ratio (annualized)	0.72	0.70	0.71
Panel C. Classification by term-spread beta			
Investors	258	57	1,772
CAPM alpha (annualized, %)	3.55	2.45	2.92
	(4.49)	(3.82)	(4.79)
Market beta	1.01	1.05	1.01
FF5 alpha (annualized, %)	2.63	2.33	2.44
	(5.92)	(4.12)	(5.40)
Sharpe ratio (annualized)	0.75	0.68	0.71

This table reports risk-adjusted performance for investors classified by the sign and significance of their first-stage multivariate macro betas. For each macro variable, investors are assigned to one of three groups: significantly positive beta, significantly negative beta, or statistically insignificant beta, using a 5 percent significance threshold. For each group and quarter, I form an equal-weighted portfolio of investor-level next-quarter raw returns. CAPM alpha is estimated from a time-series regression of quarterly excess group returns on the market factor. Fama–French five-factor alpha is estimated from a regression on the market, size, value, profitability, and investment factors. Reported alphas are annualized by multiplying quarterly alphas by four. HAC(Newey–West) t-statistics with four lags are reported in parentheses below the alpha estimates. Sharpe ratios are computed from quarterly excess returns and annualized using $\sqrt{4}$. Investor returns are constructed from reported 13F equity holdings using lagged portfolio weights and subsequent CRSP quarterly stock returns. The table therefore evaluates the performance of reported equity holdings rather than total fund returns.

Table 13: Heterogeneous Investor-Level Macro Responses by Investor Type

Macro variable	Households	Banks	Insurance	Investment advisors	Mutual funds	Pension funds	Other
GDP growth	0 (0.00%) [0,0]	24 (7.21%) [17,7]	12 (13.33%) [7,5]	204 (8.36%) [129,75]	24 (6.96%) [18,6]	6 (7.89%) [3,3]	12 (9.68%) [8,4]
Unemployment	0 (0.00%) [0,0]	27 (8.11%) [13,14]	17 (18.89%) [8,9]	332 (13.61%) [199,133]	86 (24.93%) [53,33]	9 (11.84%) [5,4]	11 (8.87%) [8,3]
Inflation	0 (0.00%) [0,0]	22 (6.61%) [8,14]	8 (8.89%) [3,5]	317 (12.99%) [168,149]	45 (13.04%) [26,19]	5 (6.58%) [1,4]	11 (8.87%) [6,5]
TERM	0 (0.00%) [0,0]	43 (12.91%) [32,11]	17 (18.89%) [10,7]	407 (16.68%) [257,150]	63 (18.26%) [37,26]	13 (17.11%) [8,5]	16 (12.90%) [11,5]
DEF	0 (0.00%) [0,0]	19 (5.71%) [9,10]	8 (8.89%) [3,5]	203 (8.32%) [105,98]	46 (13.33%) [20,26]	6 (7.89%) [6,0]	11 (8.87%) [7,4]
Wealth growth	0 (0.00%) [0,0]	31 (9.31%) [10,21]	15 (16.67%) [4,11]	320 (13.11%) [116,204]	35 (10.14%) [9,26]	10 (13.16%) [3,7]	12 (9.68%) [5,7]

Cell format is: *Significant investors (share within type) [significant positive, significant negative]*. Significance is two-sided $p < 0.05$ from investor-by-investor time-series regressions of aggregate active demand on the six macro variables.
Shares are computed using the number of investors in each type (Households N=1, Banks N=333, Insurance N=90, Investment advisors N=2440, Mutual funds N=345, Pension funds N=76, Other N=124).
The sample includes investors with at least 40 quarterly observations after merging investor-quarter aggregate active demand with macro data.

Table 14: Macro Coefficient Sign and Significance by Investor Type

Macro variable	Households	Banks	Insurance	Inv. advisors	Mutual funds	Pension funds	Other
GDP growth	0,1,0	7,309,17	5,78,7	75,2236,129	6,321,18	3,70,3	4,112,8
Unemployment	0,1,0	14,306,13	9,73,8	133,2108,199	33,259,53	4,67,5	3,113,8
Inflation	0,1,0	14,311,8	5,82,3	149,2123,168	19,300,26	4,71,1	5,113,6
TERM	0,1,0	11,290,32	7,73,10	150,2033,257	26,282,37	5,63,8	5,108,11
DEF	0,1,0	10,314,9	5,82,3	98,2237,105	26,299,20	0,70,6	4,113,7
Wealth growth	0,1,0	21,302,10	11,75,4	204,2120,116	26,310,9	7,66,3	7,112,5

Each cell reports a triplet X, Y, Z: X = number of investors with a significantly negative coefficient ($p < 0.05$); Y = number of investors with a non-significant coefficient ($p \geq 0.05$); Z = number of investors with a significantly positive coefficient ($p < 0.05$).
Regressions are estimated separately for each investor (heterogeneous time-series model) using aggregate active demand as the dependent variable and six macro variables as regressors.
Sample includes investors with at least 40 quarterly observations after merging with macro data. Investor-type columns are based on the project's manager-type coding.

Table 15: Type-Level Aggregate Active Demand Regressions on Six Macroeconomic Variables

	Households	Banks	Insurance	Inv. advisors	Mutual funds	Pension funds	Other
GDP growth	-8560.401*	343.822	130.185	-480.143	-318.414	289.531	413.202
Unemployment	9856.783	-93.254	-1186.066	-709.401	-1761.355	124.618	368.592
Inflation	7942.211	-1770.152	-407.547	691.798	-4417.335	78.162	-206.309
TERM	-395.490	705.319	-385.070	3425.786*	-5876.046	1042.559**	-1154.500
DEF	-57301.452	217.115	-92.542	-3928.774	-24421.920	2796.946	-1581.924
Wealth growth	2353.515	490.650	-0.939	34.836	-208.355	58.105	-114.749
Observations	178	178	178	178	178	178	178
R^2	0.034	0.039	0.046	0.027	0.040	0.086	0.017

This table reports one quarterly time-series regression per investor type, where the dependent variable is type-level aggregate active demand (sum of investor-level aggregate active demand within type at each quarter). Regressors are GDP growth, unemployment, inflation, TERM, DEF, and wealth growth. The estimation window has 178 quarters for each type.
Coefficients are shown. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.
The left-hand-side series is winsorized at the 1st/99th percentiles within type before estimation. Macro series are the same six transformed quarterly variables used in the heterogeneous investor-level analysis.
Interpretation: each coefficient is the change in type-level aggregate active demand (in dollars) associated with a one-unit change in the corresponding macro variable.

7 Appendix E

8 Performance Evaluation

Investor-level portfolio returns are constructed as value-weighted returns on reported equity holdings using lagged portfolio weights and CRSP stock returns. For each investor i, the next-quarter return is given by

$$R_{i,t+1} = \sum_j w_{i,j,t} r_{j,t+1},$$

where $w_{i,j,t}$ denotes the weight of stock j in investor i's portfolio at the end of quarter t, based on market values, and $r_{j,t+1}$ is the realized return over quarter $t+1$. Returns include dividends, and delisting-adjusted returns are used when necessary.

To evaluate performance over different horizons, I construct cumulative returns by compounding quarterly returns. Specifically, one-year and five-year returns are obtained by compounding over four and twenty quarters, respectively. For the five-year horizon, I also report annualized returns. This construction ensures that all returns are based on information available at the beginning of each period and correspond to standard buy-and-hold performance.

8.1 Performance by Macro Cyclicality

To assess whether macro-sensitive investors earn systematically different future returns, I classify investors according to the sign and statistical significance of their macro betas. For each macro variable—dividend yield, default premium, and term spread—investors are grouped into positive-beta, negative-beta, and insignificant-beta categories based on the first-stage multivariate regressions. For each group, I compute equal-weighted average future returns across investors.

The results show that macro cyclicality is associated with economically meaningful dif-

ferences in future performance, although patterns vary across risk-premium proxies. For dividend yield, investors with positive betas—those that increase equity exposure when expected returns are high—earn higher future returns across all horizons, with the performance gap widening at longer horizons. This pattern is consistent with long-horizon institutions absorbing risk when compensation for bearing it is elevated and subsequently earning higher realized returns.

For the default premium, performance differences are smaller and more symmetric. Both positive- and negative-beta investors perform similarly, with only modest differences relative to the insignificant group. This reflects the weaker and less directional predictive content of credit conditions over the sample period.

In contrast, term-spread sensitivities display a different pattern. Investors with negative betas—those that reduce exposure when the yield curve steepens—earn higher future returns. This suggests that responses to term-structure variation capture a distinct dimension of risk-taking behavior, potentially related to growth expectations, duration risk, or the timing of business-cycle turning points.

Overall, these findings indicate that institutional responses to macroeconomic conditions are systematically related to future performance, but the direction and strength of this relationship depend on the underlying source of variation in expected returns.

8.2 Risk-Adjusted Performance

I next examine whether these performance differences persist after adjusting for risk. For each macro-cyclicality group, I form portfolios of investor-level returns and evaluate their performance using CAPM and Fama–French five-factor models, as well as Sharpe ratios.

Across all classifications, market betas are close to one, indicating that these portfolios have aggregate exposure comparable to the broad equity market. Estimated alphas are positive and statistically significant, reflecting economically meaningful abnormal returns within the reported 13F equity portfolios. The positive-beta groups for dividend yield and

term spread exhibit somewhat stronger risk-adjusted performance than the corresponding negative- and insignificant-beta groups, while differences across default-premium groups are modest.

These results confirm that macro-driven differences in equity exposure are not only associated with variation in raw returns, but also translate into differences in risk-adjusted performance. Taken together, the evidence suggests that heterogeneity in institutional responses to business-cycle conditions has meaningful implications for both realized returns and the distribution of risk-adjusted performance across investors.

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